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(54) **ϵ -VOPO₄ CATHODE FOR LITHIUM ION BATTERIES**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,355,162 A 10/1982 Johnson

4,376,709 A 3/1983 Johnson

(Continued)

OTHER PUBLICATIONS

Berrah, Fadila, et al. "The vanadium monophosphates AVOPO₄: Synthesis of a second form β -KVOPO₄ and structural relationships in the series." Solid state sciences 3.4 (2001): 477-482.

(Continued)

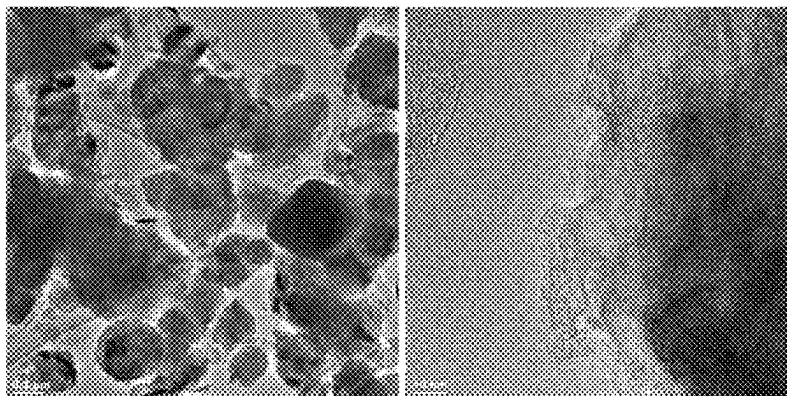
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(57) **ABSTRACT**

The epsilon polymorph of vanadyl phosphate, ϵ -VOPO₄, made from the solvothermally synthesized H₂VOPO₄, is a high-density cathode material for lithium-ion batteries optimized to reversibly intercalate two Li-ions to reach the full theoretical capacity at least 50 cycles with a coulombic efficiency of 98%. This material adopts a stable 3D tunnel structure and can extract two Li-ions per vanadium ion, giving a theoretical capacity of 305 mAh/g, with an upper charge/discharge plateau at around 4.0 V, and one lower at around 2.5 V.

20 Claims, 6 Drawing Sheets



TEM images of ϵ -VOPO₄ hand ground with graphene nanoplatelets for
electrode preparation

Related U.S. Application Data

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,459,406 A 7/1984 Johnson
4,465,743 A 8/1984 Skarstad
4,619,874 A 10/1986 Skarstad et al.
4,744,787 A 5/1988 Phipps et al.
5,061,581 A 10/1991 Narang et al.
5,114,809 A 5/1992 Nakacho et al.
5,135,477 A 8/1992 Untereker et al.
5,153,080 A 10/1992 Inubushi et al.
5,415,628 A 5/1995 Untereker et al.
5,419,890 A 5/1995 Saidi
5,437,692 A 8/1995 Dasgupta et al.
5,443,809 A 8/1995 Olsen
5,453,261 A 9/1995 Saidi et al.
5,482,697 A 1/1996 Saidi
5,498,489 A 3/1996 Dasgupta et al.
5,512,389 A 4/1996 Dasgupta et al.
5,573,503 A 11/1996 Untereker et al.
5,576,120 A 11/1996 Saidi
5,599,643 A 2/1997 Plichta et al.
5,639,577 A 6/1997 Takeuchi et al.
5,667,916 A 9/1997 Ebel et al.
5,711,761 A 1/1998 Untereker et al.
5,731,104 A 3/1998 Ventura et al.
5,759,715 A 6/1998 Barker et al.
5,849,434 A 12/1998 Miura et al.
5,851,696 A 12/1998 Saidi et al.
6,007,945 A 12/1999 Jacobs et al.
6,015,638 A 1/2000 Ventura et al.
6,124,057 A 9/2000 Ito et al.
6,156,395 A 12/2000 Zhang et al.
6,225,007 B1 5/2001 Horne et al.
6,331,282 B1 12/2001 Manthiram et al.
6,337,155 B1 1/2002 Takei et al.
6,361,755 B1 3/2002 Manthiram et al.
6,391,494 B2 5/2002 Reitz et al.
6,403,253 B1 6/2002 Wainwright et al.
6,492,061 B1 12/2002 Gauthier et al.
6,503,646 B1 1/2003 Ghantous et al.
6,528,033 B1 3/2003 Barker et al.
6,645,452 B1 11/2003 Barker et al.
6,653,022 B2 11/2003 Nordlinder et al.
6,702,961 B2 3/2004 Barker et al.
6,716,372 B2 4/2004 Barker et al.
6,723,470 B2 4/2004 Barker et al.
6,730,437 B2 5/2004 Leising et al.
6,749,966 B2 6/2004 Reitz et al.

6,805,999 B2 10/2004 Lee et al.
6,838,023 B2 1/2005 Torardi
6,844,047 B2 1/2005 Kaminsky et al.
6,855,273 B2 2/2005 Ravet et al.
6,859,297 B2 2/2005 Lee et al.
6,872,492 B2 3/2005 Barker et al.
6,884,544 B2 4/2005 Barker et al.
6,888,663 B2 5/2005 Bourdelais et al.
6,913,855 B2 7/2005 Stoker et al.
6,958,860 B2 10/2005 Dontula et al.
6,960,331 B2 11/2005 Barker et al.
6,962,666 B2 11/2005 Ravet et al.
7,001,690 B2 2/2006 Barker et al.
7,026,072 B2 4/2006 Barker et al.
7,041,239 B2 5/2006 Barker et al.
7,045,249 B2 5/2006 Nanjundaswamy et al.
7,192,564 B2 3/2007 Cardarelli et al.
7,198,869 B2 4/2007 Ghantous et al.
7,300,722 B2 11/2007 Christian et al.
7,338,734 B2 3/2008 Chiang et al.
7,344,659 B2 3/2008 Ravet et al.
7,348,100 B2 3/2008 Adamson et al.
7,383,083 B2 6/2008 Fischer et al.
7,422,823 B2 9/2008 Saidi et al.
7,438,992 B2 10/2008 Barker et al.
7,482,097 B2 1/2009 Saidi et al.
7,632,602 B2 12/2009 Lee et al.
7,722,787 B2 5/2010 Horne et al.
7,759,008 B2 7/2010 Barker et al.
7,771,628 B2 8/2010 Barker et al.
7,790,319 B2 9/2010 Faulkner
7,815,819 B2 10/2010 Ravet et al.
7,901,810 B2 3/2011 Barker et al.
7,906,239 B2 3/2011 Watanabe et al.
7,927,742 B2 4/2011 Scott et al.
7,939,201 B2 5/2011 Chiang et al.
7,952,090 B2 5/2011 Kugler
7,955,733 B2 6/2011 Armand et al.
7,960,058 B2 6/2011 Goodenough et al.
7,960,059 B2 6/2011 Adamson et al.
7,964,308 B2 6/2011 Goodenough et al.
7,972,728 B2 7/2011 Goodenough et al.
7,988,746 B2 8/2011 Chiang et al.
7,998,617 B2 8/2011 Armand et al.
8,003,250 B2 8/2011 Chiang et al.
8,057,936 B2 11/2011 Chiang et al.
8,067,117 B2 11/2011 Goodenough et al.
8,088,512 B2 1/2012 Gozdz et al.
8,119,285 B2 2/2012 Hatta et al.
8,148,009 B2 4/2012 Chiang et al.
8,148,013 B2 4/2012 Chiang et al.
8,158,090 B2 4/2012 Chiang et al.
8,168,326 B2 5/2012 Chiang et al.
8,173,049 B2 5/2012 Ravet et al.
8,178,239 B2 5/2012 Hatta et al.
8,206,468 B2 6/2012 Chiang et al.
8,206,469 B2 6/2012 Chiang et al.
8,218,351 B2 7/2012 Bloch et al.
8,241,789 B2 8/2012 Chiang et al.
8,241,792 B2 8/2012 Kaneko et al.
8,257,616 B2 9/2012 Ravet et al.
8,277,975 B2 10/2012 Chiang et al.
8,282,691 B2 10/2012 Goodenough et al.
8,309,242 B2 11/2012 Wei et al.
8,318,352 B2 11/2012 Saidi et al.
8,318,353 B2 11/2012 Faulkner
8,323,831 B2 12/2012 Wilkins et al.
8,323,832 B2 12/2012 Chiang et al.
8,329,071 B2 12/2012 Wang et al.
8,337,723 B2 12/2012 Hashiba et al.
8,361,661 B2 1/2013 Doe et al.
8,383,269 B2 2/2013 Scott et al.
8,420,258 B2 4/2013 Rojeski
8,435,678 B2 5/2013 Wixom et al.
8,435,738 B2 5/2013 Holmes
8,445,135 B2 5/2013 Sano et al.
8,449,980 B2 5/2013 Suzuki et al.
8,475,739 B2 7/2013 Holmes et al.
8,481,208 B2 7/2013 Riley, Jr. et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,491,861 B2	7/2013	Evenson et al.	9,337,481 B2	5/2016	Nesper et al.
8,492,031 B2	7/2013	Suzuki et al.	9,349,544 B2	5/2016	Rojeski
8,506,847 B2	8/2013	Hibst et al.	9,350,018 B2	5/2016	Tsutsumi et al.
8,506,851 B2	8/2013	Ravet et al.	9,362,555 B2	6/2016	Chen et al.
8,506,852 B2	8/2013	Ravet et al.	9,362,562 B2	6/2016	Armand et al.
8,580,430 B2	11/2013	Chiang et al.	9,368,831 B2	6/2016	He et al.
8,586,238 B2	11/2013	Chiang et al.	9,385,364 B2	7/2016	Liang et al.
8,614,020 B2	12/2013	Kondo et al.	9,391,290 B2	7/2016	Watabe et al.
8,617,430 B2	12/2013	Chiang et al.	9,401,528 B2	7/2016	Doe et al.
8,617,745 B2	12/2013	Gozdz et al.	9,406,930 B2	8/2016	Fukui et al.
8,632,904 B2	1/2014	Guyomard et al.	9,413,006 B2	8/2016	Kay
8,652,683 B2	2/2014	Rojeski	9,419,279 B2	8/2016	Yamamura et al.
8,658,310 B2	2/2014	Rojeski	9,437,866 B2	9/2016	Kikuchi et al.
8,709,647 B2	4/2014	Gozdz et al.	9,449,765 B2	9/2016	Gardner et al.
8,722,227 B2	5/2014	Chiang et al.	9,450,209 B2	9/2016	Seo et al.
8,722,242 B2	5/2014	Muldoon et al.	9,461,092 B2	10/2016	Seo et al.
8,734,539 B2	5/2014	Tokita et al.	9,472,601 B2	10/2016	Seo et al.
8,784,694 B2	7/2014	Kay	9,515,310 B2	12/2016	Cao et al.
8,785,043 B2	7/2014	Armand et al.	9,525,164 B1	12/2016	Aksu et al.
8,798,235 B2	8/2014	Ohta et al.	9,527,157 B2	12/2016	Kroll et al.
8,798,236 B2	8/2014	Ohta et al.	9,537,109 B2	1/2017	Suzuki et al.
8,802,295 B2	8/2014	Nesper et al.	9,548,509 B2	1/2017	Anderson et al.
8,808,575 B2	8/2014	Hibst et al.	9,553,312 B2	1/2017	Yamauchi et al.
8,808,576 B2	8/2014	Hibst et al.	9,559,351 B2	1/2017	Mori et al.
8,821,763 B2	9/2014	Sano et al.	9,564,641 B2	2/2017	Otsuki et al.
8,822,080 B2	9/2014	Fujino et al.	9,581,877 B2	2/2017	Bass et al.
8,840,838 B2	9/2014	Holmes et al.	9,583,764 B2	2/2017	Yamauchi et al.
8,852,807 B2	10/2014	Chiang et al.	9,592,508 B2	3/2017	Holmes et al.
8,870,976 B2	10/2014	Kondo et al.	9,595,697 B2	3/2017	Nishido et al.
8,895,193 B2	11/2014	Abouimrane et al.	9,608,269 B2	3/2017	Barker
8,926,389 B2	1/2015	Nishido	9,627,691 B2	4/2017	Xing et al.
8,929,510 B2	1/2015	Nishino et al.	9,632,102 B2	4/2017	Holmes et al.
8,932,762 B2	1/2015	Sano et al.	9,634,267 B2	4/2017	Watabe et al.
8,936,871 B2	1/2015	Sano et al.	9,645,143 B2	5/2017	Holmes et al.
8,951,676 B2	2/2015	Doe et al.	9,653,517 B2	5/2017	Uesaka et al.
8,951,680 B2	2/2015	Doe et al.	9,653,730 B2	5/2017	Pitteloud
8,957,442 B2	2/2015	Seo et al.	9,664,702 B2	5/2017	Holmes et al.
8,980,453 B2	3/2015	Scott et al.	9,666,755 B2	5/2017	Shimomura
8,993,171 B2	3/2015	Sano et al.	9,683,314 B2	6/2017	Xu et al.
8,999,282 B2	4/2015	Ceder et al.	9,685,656 B2	6/2017	Yamauchi et al.
8,999,571 B2	4/2015	Chiang et al.	9,711,793 B2	7/2017	Mizuno et al.
9,000,458 B2	4/2015	Seo et al.	9,719,990 B2	8/2017	Holmes et al.
9,006,755 B2	4/2015	Seo et al.	9,722,247 B2	8/2017	Zhang et al.
9,029,003 B2	5/2015	Sulaiman Lo et al.	9,745,194 B2	8/2017	Kabe et al.
9,044,191 B2	6/2015	Nishino et al.	9,771,319 B2	9/2017	Gonzalez Martinez et al.
9,065,093 B2	6/2015	Chiang et al.	9,786,913 B2	10/2017	Inoue et al.
9,077,032 B2	7/2015	Doe et al.	9,786,944 B2	10/2017	Chiang et al.
9,077,037 B2	7/2015	Hwu et al.	9,793,061 B2	10/2017	Gardner et al.
9,093,707 B2	7/2015	Lee et al.	9,798,214 B2	10/2017	Garcia et al.
9,105,869 B2	8/2015	Nishido	9,810,704 B2	11/2017	Holmes et al.
9,112,210 B2	8/2015	Chen et al.	9,819,054 B2	11/2017	Kang et al.
9,128,015 B2	9/2015	Holmes et al.	9,853,242 B2	12/2017	Nishido
9,136,505 B2	9/2015	Sasaki et al.	9,859,557 B2	1/2018	Mori et al.
9,147,874 B2	9/2015	Chen et al.	9,865,877 B2	1/2018	Kang et al.
9,153,839 B2	10/2015	Furuya	9,878,305 B2	1/2018	Hossain et al.
9,168,016 B2	10/2015	Ohta et al.	9,882,204 B2	1/2018	Ryoshi et al.
9,172,111 B2	10/2015	Doe et al.	9,887,047 B2	2/2018	Tan et al.
9,190,659 B2	11/2015	Kondo et al.	9,899,672 B2	2/2018	Zhamu et al.
9,196,923 B2	11/2015	Chang et al.	9,899,710 B2	2/2018	Scott et al.
9,203,084 B2	12/2015	Wang et al.	9,911,985 B2	3/2018	Dong et al.
9,240,612 B2	1/2016	Doe et al.	9,917,271 B2	3/2018	Uesaka et al.
9,250,229 B2	2/2016	Holmes	9,923,206 B2	3/2018	Chen et al.
9,257,655 B2	2/2016	Suzuki et al.	9,929,368 B2	3/2018	Kataishi et al.
9,257,718 B2	2/2016	Kondo et al.	9,935,158 B2	4/2018	Seo et al.
9,258,853 B2	2/2016	Nishido et al.	9,935,286 B2	4/2018	Watabe et al.
9,268,915 B2	2/2016	Holmes et al.	9,935,314 B2	4/2018	Johnson et al.
9,287,578 B2	3/2016	Anderson et al.	9,942,972 B2	4/2018	Nishino et al.
9,287,580 B2	3/2016	Jain	9,952,240 B2	4/2018	Holmes et al.
9,293,790 B2	3/2016	Doe et al.	9,954,224 B2	4/2018	Toya et al.
9,296,621 B2	3/2016	Feng et al.	9,960,451 B1	5/2018	Zhamu et al.
9,299,966 B2	3/2016	Chiang et al.	9,966,604 B2	5/2018	Kondo et al.
9,305,716 B2	4/2016	Tan et al.	9,997,778 B2	6/2018	Cao et al.
9,318,739 B2	4/2016	Toya et al.	10,007,163 B2	6/2018	Bass et al.
9,318,742 B2	4/2016	Fujino et al.	10,008,723 B1	6/2018	Zhamu et al.
9,331,283 B2	5/2016	Lim et al.	10,008,747 B1	6/2018	Zhamu et al.
			10,012,664 B2	7/2018	Wasson et al.
			10,017,875 B2	7/2018	Toya et al.
			10,018,643 B2	7/2018	Holmes et al.
			10,044,025 B2	8/2018	Toya et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,050,271 B2	8/2018	Barker et al.	10,734,648 B2	8/2020	Zhamu et al.
10,069,097 B2	9/2018	Kataishi et al.	10,734,671 B2	8/2020	Zhamu et al.
10,069,138 B2	9/2018	Mizuno et al.	10,741,619 B2	8/2020	Seo et al.
10,084,156 B2	9/2018	Seo et al.	10,741,832 B2	8/2020	Sun et al.
10,090,516 B2	10/2018	Balaya et al.	10,749,168 B1	8/2020	Johnson et al.
10,096,658 B2	10/2018	Watabe et al.	10,756,286 B2	8/2020	Ishisone et al.
10,109,849 B2	10/2018	Ryoshi et al.	10,763,491 B2	9/2020	Takeuchi et al.
10,109,859 B1	10/2018	Goodenough et al.	10,766,795 B2	9/2020	Su et al.
10,128,501 B2	11/2018	Fukui et al.	10,770,721 B2	9/2020	Zhamu et al.
10,130,936 B2	11/2018	Hossain et al.	10,772,199 B2	9/2020	Lai et al.
10,134,998 B2	11/2018	Seo et al.	10,777,810 B2	9/2020	He et al.
10,153,516 B2	12/2018	Schmitz et al.	10,784,509 B2	9/2020	He et al.
10,160,660 B1	12/2018	Rodriguez et al.	10,790,509 B2	9/2020	Uekusa et al.
10,170,212 B2	1/2019	Barker et al.	10,797,113 B2	10/2020	Ohsawa et al.
10,170,749 B2	1/2019	Zhamu et al.	10,797,313 B2	10/2020	Zhamu et al.
10,170,789 B2	1/2019	Zhamu et al.	10,818,921 B2	10/2020	Kensaku et al.
10,199,637 B2	2/2019	Zhamu et al.	10,826,132 B2	11/2020	Smith et al.
10,199,647 B2	2/2019	Chen et al.	10,829,385 B2	11/2020	Matsumoto et al.
10,201,065 B2	2/2019	Ohta et al.	10,840,501 B2	11/2020	Hong et al.
10,207,992 B2	2/2019	Suzuki et al.	10,840,505 B2	11/2020	Zarra et al.
10,224,494 B2	3/2019	Watabe et al.	10,840,565 B2	11/2020	Zhamu et al.
10,230,055 B2	3/2019	Kanamoto et al.	10,854,396 B2	12/2020	Yokota et al.
10,230,108 B2	3/2019	Sano et al.	10,862,129 B2	12/2020	Pan et al.
10,243,217 B2	3/2019	Zhamu et al.	10,868,256 B2	12/2020	Seo et al.
10,263,253 B2	4/2019	Nair et al.	10,873,083 B2	12/2020	Zhamu et al.
10,263,277 B2	4/2019	Pitts et al.	10,886,525 B2	1/2021	Singh et al.
10,263,308 B2	4/2019	McKone et al.	10,886,528 B2	1/2021	Pan et al.
10,270,098 B2	4/2019	Kang et al.	10,889,506 B2	1/2021	Rodriguez et al.
10,276,856 B2	4/2019	Zhamu et al.	10,903,519 B2	1/2021	Pan et al.
10,333,138 B2	6/2019	Goodenough et al.	10,903,527 B2	1/2021	Liu et al.
10,340,470 B2	7/2019	Sasaki et al.	10,910,568 B2	2/2021	Watabe et al.
10,361,388 B2	7/2019	Mitsumori et al.	10,910,576 B2	2/2021	Mitsumori et al.
10,361,460 B2	7/2019	Zhamu et al.	10,916,769 B2	2/2021	Su et al.
10,371,710 B2	8/2019	Wasson et al.	10,923,717 B2	2/2021	Kim et al.
10,388,983 B2	8/2019	Zhamu et al.	10,930,924 B2	2/2021	Zhamu et al.
10,396,356 B2	8/2019	Toya et al.	10,937,965 B2	3/2021	Seo et al.
10,411,264 B2	9/2019	Pan et al.	2001/0046468 A1	11/2001	Reitz et al.
10,418,662 B2	9/2019	Zhamu et al.	2002/0039687 A1	4/2002	Barker et al.
10,424,787 B2	9/2019	Kamata et al.	2002/0064497 A1	5/2002	Horne et al.
10,446,845 B2	10/2019	Goodenough et al.	2002/0086212 A1	7/2002	Nordlinder et al.
10,446,853 B2	10/2019	Choi et al.	2002/0086214 A1	7/2002	Barker et al.
10,461,134 B2	10/2019	Seo et al.	2002/0142218 A1	10/2002	Reitz et al.
10,483,533 B2	11/2019	Zhamu et al.	2002/0192553 A1	12/2002	Barker et al.
10,490,815 B2	11/2019	Toya et al.	2002/0195591 A1	12/2002	Ravet et al.
10,493,432 B2	12/2019	Islam et al.	2003/0003362 A1	1/2003	Leising et al.
10,516,165 B2	12/2019	Toya et al.	2003/0022065 A1	1/2003	Zhang et al.
10,518,265 B2	12/2019	Holmes et al.	2003/0073003 A1	4/2003	Barker et al.
10,534,009 B2	1/2020	Holmes et al.	2003/0077513 A1	4/2003	Ghantous et al.
10,535,880 B2	1/2020	Zhamu et al.	2003/0077514 A1	4/2003	Barker et al.
10,535,892 B2	1/2020	Zhamu et al.	2003/0129492 A1	7/2003	Barker et al.
10,553,857 B2	2/2020	Mizuno et al.	2003/0162094 A1	8/2003	Lee et al.
10,557,863 B2	2/2020	Holmes et al.	2003/0190526 A1	10/2003	Saidi et al.
10,559,815 B2	2/2020	Zhamu et al.	2003/0190527 A1	10/2003	Pugh et al.
10,573,932 B2	2/2020	Yang et al.	2003/0190528 A1	10/2003	Saidi et al.
10,586,931 B2	3/2020	Kanamoto et al.	2003/0215714 A1	11/2003	Barker et al.
10,611,639 B2	4/2020	Sano et al.	2003/0235761 A1	12/2003	Prabaharan
10,615,408 B1	4/2020	Fellner et al.	2004/0005265 A1	1/2004	Chiang et al.
10,615,453 B2	4/2020	Zanotti et al.	2004/0005500 A1	1/2004	Torardi
10,615,463 B2	4/2020	Scott et al.	2004/0013943 A1	1/2004	Stoker et al.
10,627,418 B2	4/2020	Wasson et al.	2004/0016632 A1	1/2004	Barker
10,637,043 B2	4/2020	Zhamu et al.	2004/0028585 A1	2/2004	Cardarelli et al.
10,637,115 B2	4/2020	Licht	2004/0048157 A1	3/2004	Neudecker et al.
10,644,250 B2	5/2020	Watabe et al.	2004/0066556 A1	4/2004	Dontula et al.
10,644,308 B2	5/2020	Liu et al.	2004/0121195 A1	6/2004	Ghantous et al.
10,658,604 B2	5/2020	Mitsumori et al.	2004/0126300 A1	7/2004	Barker et al.
10,661,090 B2	5/2020	Chen et al.	2004/0140458 A1	7/2004	Ravet et al.
10,669,646 B2	6/2020	Toya et al.	2004/0150867 A1	8/2004	Lee et al.
10,673,090 B2	6/2020	Nie et al.	2004/0197654 A1	10/2004	Barker et al.
10,693,094 B2	6/2020	Seo et al.	2004/0262571 A1	12/2004	Barker et al.
10,697,082 B1	6/2020	Huang et al.	2005/0003269 A1	1/2005	Nanjundaswamy et al.
10,698,286 B2	6/2020	Bass et al.	2005/0181283 A1	8/2005	Pugh et al.
10,727,479 B2	7/2020	Zhamu et al.	2005/0186476 A1	8/2005	Barker et al.
10,727,531 B2	7/2020	Zhamu et al.	2005/0194567 A1	9/2005	Adamson et al.
10,734,635 B2	8/2020	Zhamu et al.	2005/0238961 A1	10/2005	Barker et al.
10,734,646 B2	8/2020	He et al.	2006/0060827 A1	3/2006	Ravet et al.
			2006/0088767 A1	4/2006	Li et al.
			2006/0110580 A1	5/2006	Aylward et al.
			2006/0194112 A1	8/2006	Barker et al.
			2006/0228629 A1	10/2006	Christian et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0292444	A1	12/2006	Chiang et al.	2012/0164499	A1	6/2012	Chiang et al.
2007/0009802	A1	1/2007	Lee et al.	2012/0164517	A1	6/2012	Lee et al.
2007/0031732	A1	2/2007	Chiang et al.	2012/0205676	A1	8/2012	Seo et al.
2007/0031735	A1	2/2007	Nishimura et al.	2012/0205685	A1	8/2012	Seo et al.
2007/0065724	A1	3/2007	Barker et al.	2012/0205686	A1	8/2012	Seo et al.
2007/0072034	A1	3/2007	Barker et al.	2012/0213919	A1	8/2012	Ravet et al.
2007/0082262	A1	4/2007	Nishimura et al.	2012/0214071	A1	8/2012	Chiang et al.
2007/0141468	A1	6/2007	Barker	2012/0214072	A1	8/2012	Ravet et al.
2007/0166617	A1	7/2007	Gozdz et al.	2012/0219856	A1	8/2012	Doe et al.
2007/0190418	A1	8/2007	Chiang et al.	2012/0219859	A1	8/2012	Doe et al.
2007/0248520	A1	10/2007	Faulkner	2012/0219862	A1	8/2012	Fujino et al.
2007/0292747	A1	12/2007	Chiang et al.	2012/0237828	A1	9/2012	Tan et al.
2007/0292757	A1	12/2007	Watanabe et al.	2012/0270107	A1	10/2012	Toya et al.
2007/0298317	A1	12/2007	Brodd et al.	2012/0270109	A1	10/2012	Chiang et al.
2008/0014503	A1	1/2008	Kejha	2012/0276454	A1	11/2012	Mori et al.
2008/0014507	A1	1/2008	Kejha	2012/0301787	A1	11/2012	Yamamura et al.
2008/0052900	A1	3/2008	Kondo et al.	2012/0302697	A1	11/2012	Inada et al.
2008/0057390	A1	3/2008	Kondo et al.	2012/0315538	A1	12/2012	Chiang et al.
2008/0070113	A1	3/2008	Kondo et al.	2012/0321953	A1*	12/2012	Chen H01M 4/483 977/773
2008/0131777	A1	6/2008	Hatta et al.	2013/0026409	A1	1/2013	Baker et al.
2008/0138709	A1	6/2008	Hatta et al.	2013/0029207	A1	1/2013	Jain
2008/0157024	A1	7/2008	Adamson et al.	2013/0034780	A1	2/2013	Muldoon et al.
2008/0257721	A1	10/2008	Ravet et al.	2013/0043437	A1	2/2013	Wang et al.
2008/0261113	A1	10/2008	Huang et al.	2013/0048967	A1	2/2013	Nishido et al.
2009/0040587	A1	2/2009	Kugler	2013/0052492	A1	2/2013	Schaefer
2009/0061314	A1	3/2009	Dong	2013/0059204	A1	3/2013	Holzappel et al.
2009/0081553	A1	3/2009	Kondo et al.	2013/0059211	A1	3/2013	Schaefer et al.
2009/0214944	A1	8/2009	Rojeski	2013/0074614	A1	3/2013	Holmes et al.
2009/0220838	A9	9/2009	Barker et al.	2013/0078149	A1	3/2013	Holmes et al.
2009/0246636	A1	10/2009	Chiang et al.	2013/0078520	A1	3/2013	Toya et al.
2009/0253036	A1	10/2009	Owen et al.	2013/0078523	A1	3/2013	Rojeski
2009/0272949	A1	11/2009	Buttry	2013/0078524	A1	3/2013	Rojeski
2009/0303660	A1	12/2009	Nair et al.	2013/0078624	A1	3/2013	Holmes et al.
2009/0311597	A1	12/2009	Chiang et al.	2013/0078625	A1	3/2013	Holmes et al.
2009/0311680	A1	12/2009	Nakao et al.	2013/0078733	A1	3/2013	Holmes et al.
2010/0055573	A1	3/2010	Lee et al.	2013/0079236	A1	3/2013	Holmes
2010/0075225	A1	3/2010	Wilkins et al.	2013/0079599	A1	3/2013	Holmes et al.
2010/0078591	A1	4/2010	Sano et al.	2013/0084474	A1	4/2013	Mills
2010/0119951	A1	5/2010	Abouimrane et al.	2013/0089486	A1	4/2013	Ceder et al.
2010/0140560	A1	6/2010	Wang et al.	2013/0108925	A1	5/2013	Holzappel
2010/0143769	A1	6/2010	Lee et al.	2013/0115521	A1	5/2013	Doe et al.
2010/0230632	A1	9/2010	Adamson et al.	2013/0119358	A1	5/2013	Sasaki et al.
2010/0233545	A1	9/2010	Sano et al.	2013/0137200	A1	5/2013	Shimomura
2010/0266899	A1	10/2010	Barker et al.	2013/0153821	A1	6/2013	Evenson et al.
2010/0283012	A1	11/2010	Hibst et al.	2013/0157126	A1	6/2013	Sulaiman Lo et al.
2010/0285372	A1	11/2010	Lee et al.	2013/0157135	A1	6/2013	Zhou et al.
2010/0301281	A1	12/2010	Hibst et al.	2013/0189590	A1	7/2013	Jean-Marc et al.
2010/0303701	A1	12/2010	Faulkner	2013/0214201	A1	8/2013	Hibst et al.
2011/0008678	A1	1/2011	Li et al.	2013/0214212	A1	8/2013	Hibst et al.
2011/0012067	A1	1/2011	Kay	2013/0214462	A1	8/2013	Kikuchi et al.
2011/0052473	A1	3/2011	Sano et al.	2013/0216903	A1	8/2013	Pitteloud
2011/0052986	A1	3/2011	Barker et al.	2013/0252112	A1	9/2013	Doe et al.
2011/0052995	A1	3/2011	Sano et al.	2013/0252114	A1	9/2013	Doe et al.
2011/0084238	A1	4/2011	Hibst et al.	2013/0260225	A1	10/2013	Doe et al.
2011/0086273	A1	4/2011	Ravet et al.	2013/0260228	A1	10/2013	Sano et al.
2011/0097479	A1	4/2011	Ravet et al.	2013/0260238	A1	10/2013	Doe et al.
2011/0104534	A1	5/2011	Wei et al.	2013/0273402	A1	10/2013	Tsutsumi et al.
2011/0117417	A1	5/2011	Pitts	2013/0288129	A1	10/2013	Toya et al.
2011/0123866	A1	5/2011	Pan et al.	2013/0300284	A1	11/2013	Nishido
2011/0159381	A1	6/2011	Doe et al.	2013/0302673	A1	11/2013	Ravet et al.
2011/0186741	A1	8/2011	Ohta et al.	2013/0302697	A1	11/2013	Wang et al.
2011/0188630	A1	8/2011	Ohta et al.	2013/0309561	A1	11/2013	Chen et al.
2011/0188633	A1	8/2011	Ohta et al.	2013/0316250	A1	11/2013	Khalifah et al.
2011/0195306	A1	8/2011	Chiang et al.	2013/0323608	A1	12/2013	Chang et al.
2011/0200848	A1	8/2011	Chiang et al.	2013/0327249	A1	12/2013	Winter et al.
2011/0229765	A1	9/2011	Barker et al.	2013/0330603	A1	12/2013	Kroll et al.
2011/0274948	A1	11/2011	Duduta et al.	2013/0330611	A1	12/2013	Chen et al.
2011/0287316	A1	11/2011	Lu et al.	2013/0337331	A1	12/2013	Mori et al.
2011/0317817	A1	12/2011	Nishino et al.	2013/0344367	A1	12/2013	Chiang et al.
2011/0318652	A1	12/2011	Furuya	2014/0011090	A1	1/2014	Toya et al.
2012/0002784	A1	1/2012	Nishino et al.	2014/0027743	A1	1/2014	Nishido
2012/0064401	A1	3/2012	Liu et al.	2014/0056797	A1	2/2014	Kabe et al.
2012/0100402	A1	4/2012	Nesper et al.	2014/0072871	A1	3/2014	Chen et al.
2012/0138867	A1	6/2012	Liang et al.	2014/0072879	A1	3/2014	Chen et al.
2012/0140378	A1	6/2012	Tan et al.	2014/0072883	A1	3/2014	Kondo et al.
				2014/0073043	A1	3/2014	Holmes
				2014/0099557	A1	4/2014	Doe et al.
				2014/0106214	A1	4/2014	Doe et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2014/0127562 A1	5/2014	Nesper et al.	2016/0069920 A1	3/2016	Holmes et al.
2014/0159011 A1	6/2014	Suzuki et al.	2016/0069921 A1	3/2016	Holmes et al.
2014/0170476 A1	6/2014	Tan et al.	2016/0077015 A1	3/2016	Holmes et al.
2014/0170735 A1	6/2014	Holmes	2016/0079608 A1	3/2016	Choi et al.
2014/0186238 A1	7/2014	Holmes et al.	2016/0079609 A1	3/2016	Choi et al.
2014/0191220 A1	7/2014	Watabe et al.	2016/0084863 A1	3/2016	Holmes et al.
2014/0212733 A1	7/2014	Johnson et al.	2016/0087262 A1	3/2016	Toya et al.
2014/0220450 A1	8/2014	Jilek et al.	2016/0093885 A1	3/2016	Kamata et al.
2014/0234695 A1	8/2014	Kondo et al.	2016/0096334 A1	4/2016	Sander et al.
2014/0234949 A1	8/2014	Wasson et al.	2016/0099480 A1	4/2016	Nie et al.
2014/0242445 A1	8/2014	Gozdz et al.	2016/0103123 A1	4/2016	Holmes et al.
2014/0248544 A1	9/2014	Xing et al.	2016/0111720 A1	4/2016	Arthur et al.
2014/0275685 A1	9/2014	Sanchez Valente et al.	2016/0124009 A1	5/2016	Wasson et al.
2014/0294721 A1	10/2014	Feng et al.	2016/0155993 A1	6/2016	Nishido et al.
2014/0295275 A1	10/2014	Kay	2016/0156019 A1	6/2016	Kang et al.
2014/0296089 A1	10/2014	Holmes et al.	2016/0164093 A1	6/2016	Inoue et al.
2014/0302403 A1	10/2014	Doe et al.	2016/0164152 A1	6/2016	Tan et al.
2014/0308661 A1	10/2014	Holmes et al.	2016/0168086 A1	6/2016	Gonzalez Martinez et al.
2014/0315104 A1	10/2014	Liu et al.	2016/0169880 A1	6/2016	Holmes et al.
2014/0335415 A1	11/2014	Tamaki et al.	2016/0169923 A1	6/2016	Holmes et al.
2014/0335918 A1	11/2014	Gardner et al.	2016/0202588 A1	7/2016	Bass et al.
2014/0363746 A1	12/2014	He et al.	2016/0204436 A1	7/2016	Barker et al.
2014/0370186 A1	12/2014	Liang et al.	2016/0216286 A1	7/2016	Holmes et al.
2014/0370388 A1	12/2014	Singh et al.	2016/0216287 A1	7/2016	Holmes et al.
2014/0377660 A1	12/2014	Fukui et al.	2016/0233437 A1	8/2016	Suzuki et al.
2015/0004486 A1	1/2015	Moroishi et al.	2016/0246153 A1	8/2016	Garcia et al.
2015/0030929 A1	1/2015	Barker	2016/0268605 A1	9/2016	Yamauchi et al.
2015/0037676 A1	2/2015	Yamauchi et al.	2016/0276707 A1	9/2016	Holub et al.
2015/0044556 A1	2/2015	Wang et al.	2016/0285086 A1	9/2016	Zhu et al.
2015/0056677 A1	2/2015	Abin et al.	2016/0320381 A1	11/2016	Holmes et al.
2015/0069449 A1	3/2015	Nishido	2016/0322589 A1	11/2016	Watabe et al.
2015/0086471 A1	3/2015	Sanchez Valente et al.	2016/0343949 A1	11/2016	Seo et al.
2015/0086874 A1	3/2015	Kondo et al.	2016/0343954 A1	11/2016	Seo et al.
2015/0111105 A1	4/2015	Kato et al.	2016/0351904 A1	12/2016	Goodenough et al.
2015/0112109 A1	4/2015	Sanchez Valente et al.	2016/0358716 A1	12/2016	Gardner et al.
2015/0125945 A1	5/2015	Holmes et al.	2016/0365577 A1*	12/2016	Zhang H01M 4/5825
2015/0132650 A1	5/2015	Mizuno et al.	2016/0370396 A1	12/2016	Wasson et al.
2015/0132660 A1	5/2015	Ravet et al.	2016/0372688 A1	12/2016	Seo et al.
2015/0140476 A1	5/2015	Nair et al.	2016/0377640 A1	12/2016	Balwani et al.
2015/0155548 A1	6/2015	Ryoshi et al.	2016/0380236 A1	12/2016	Seo et al.
2015/0214273 A1	7/2015	Seo et al.	2017/0005327 A1	1/2017	Goodenough et al.
2015/0214506 A1	7/2015	Seo et al.	2017/0005332 A1	1/2017	Chen et al.
2015/0221703 A1	8/2015	Seo et al.	2017/0012207 A1	1/2017	Seo et al.
2015/0228977 A1	8/2015	Toya et al.	2017/0012232 A1	1/2017	Kataishi et al.
2015/0236342 A1	8/2015	Rojeski	2017/0025479 A1	1/2017	Seo et al.
2015/0236349 A1	8/2015	Chiang et al.	2017/0025615 A1	1/2017	Seo et al.
2015/0245456 A1	8/2015	Nishino et al.	2017/0025630 A1	1/2017	Seo et al.
2015/0263382 A1	9/2015	Singh et al.	2017/0038401 A1	2/2017	Holmes et al.
2015/0280259 A1	10/2015	Anderson et al.	2017/0040553 A1	2/2017	Watabe et al.
2015/0303474 A1	10/2015	Barker et al.	2017/0062869 A1	3/2017	Zhamu et al.
2015/0311506 A1	10/2015	Mizuno et al.	2017/0069852 A1	3/2017	Kanamoto et al.
2015/0311520 A1	10/2015	Mizuno et al.	2017/0084670 A1	3/2017	Seo et al.
2015/0311565 A1	10/2015	Muldoon et al.	2017/0092889 A1	3/2017	Seo et al.
2015/0318335 A1	11/2015	Kataishi et al.	2017/0092890 A1	3/2017	Seo et al.
2015/0338428 A1	11/2015	Holmes et al.	2017/0098856 A1	4/2017	Zhamu et al.
2015/0357653 A1	12/2015	Yoshida et al.	2017/0104204 A1	4/2017	Zhamu et al.
2015/0364753 A1	12/2015	Chiang et al.	2017/0104208 A1	4/2017	Yamauchi et al.
2015/0364761 A1	12/2015	Fukui et al.	2017/0117524 A1	4/2017	Chen et al.
2015/0368717 A1	12/2015	Holmes et al.	2017/0125703 A1	5/2017	Suzuki et al.
2015/0376817 A1	12/2015	Xu et al.	2017/0146882 A1	5/2017	Bass et al.
2016/0003823 A1	1/2016	Holmes	2017/0179527 A1	6/2017	Schmitz et al.
2016/0006028 A1	1/2016	Nair et al.	2017/0179558 A1	6/2017	McKone et al.
2016/0006090 A1	1/2016	Licht	2017/0186971 A1	6/2017	Kanamoto et al.
2016/0007432 A1	1/2016	Ohta et al.	2017/0186980 A1	6/2017	Watabe et al.
2016/0011215 A1	1/2016	Holmes	2017/0213876 A1	7/2017	Ohsawa et al.
2016/0011225 A1	1/2016	Holmes	2017/0213991 A1	7/2017	Uesaka et al.
2016/0025760 A1	1/2016	Holmes	2017/0222172 A1	8/2017	Watabe et al.
2016/0025763 A1	1/2016	Holmes	2017/0237118 A1	8/2017	Khalifah et al.
2016/0028105 A1	1/2016	Khalifah et al.	2017/0244059 A1	8/2017	Sasaki et al.
2016/0028114 A1	1/2016	Pratt et al.	2017/0256795 A1	9/2017	Semenenko et al.
2016/0032361 A1	2/2016	Holmes et al.	2017/0271610 A1	9/2017	Takahashi
2016/0033544 A1	2/2016	Holmes et al.	2017/0283845 A1	10/2017	Holmes et al.
2016/0043146 A1	2/2016	Uesaka et al.	2017/0309687 A1	10/2017	Watabe et al.
2016/0054343 A1	2/2016	Holmes et al.	2017/0309852 A1	10/2017	Seo et al.
2016/0069919 A1	3/2016	Holmes et al.	2017/0312746 A1	11/2017	Holmes et al.
			2017/0324054 A1	11/2017	Ishisone et al.
			2017/0324055 A1	11/2017	Ishisone et al.
			2017/0324081 A1	11/2017	Toya et al.
			2017/0338436 A1	11/2017	Mitsumori et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0338472	A1	11/2017	Zhamu et al.	2019/0341652	A1	11/2019	Laughman et al.
2017/0338485	A1	11/2017	Toya et al.	2019/0348625	A1	11/2019	Mitsumori et al.
2017/0350878	A1	12/2017	Holmes et al.	2019/0355963	A1	11/2019	Singh et al.
2017/0352868	A1	12/2017	Zhamu et al.	2019/0363394	A1	11/2019	Zhamu et al.
2017/0352869	A1	12/2017	Zhamu et al.	2019/0372093	A1	12/2019	Zhamu et al.
2017/0352891	A1	12/2017	Choi et al.	2019/0372099	A1	12/2019	Zhamu et al.
2017/0354955	A1	12/2017	Hossain et al.	2019/0372100	A1	12/2019	Zhamu et al.
2017/0373310	A1	12/2017	Whittingham et al.	2019/0372148	A1	12/2019	He et al.
2018/0034090	A1	2/2018	Chiang et al.	2019/0372151	A1	12/2019	Zhamu et al.
2018/0040877	A1	2/2018	Ravet et al.	2019/0372174	A1	12/2019	He et al.
2018/0045745	A1	2/2018	Holmes et al.	2019/0372449	A1	12/2019	Mills
2018/0053933	A1	2/2018	Ryoshi et al.	2019/0379021	A1	12/2019	He et al.
2018/0062214	A1	3/2018	Smith et al.	2019/0379039	A1	12/2019	Zhamu et al.
2018/0115014	A1	4/2018	Zanotti et al.	2019/0379045	A1	12/2019	He et al.
2018/0131040	A1	5/2018	Visco et al.	2019/0393467	A1	12/2019	Guo et al.
2018/0145326	A1	5/2018	Kensaku et al.	2019/0393482	A1	12/2019	He et al.
2018/0154338	A1	6/2018	Hossain et al.	2019/0393485	A1	12/2019	He et al.
2018/0159123	A1	6/2018	Mizuno et al.	2019/0393486	A1	12/2019	He et al.
2018/0169421	A1	6/2018	Chen et al.	2019/0393487	A1	12/2019	He et al.
2018/0175433	A1	6/2018	Zhamu et al.	2019/0393495	A1	12/2019	He et al.
2018/0175434	A1	6/2018	Zhamu et al.	2019/0393496	A1	12/2019	He et al.
2018/0183062	A1	6/2018	Zhamu et al.	2019/0393508	A1	12/2019	Zhamu et al.
2018/0183066	A1	6/2018	Zhamu et al.	2019/0393541	A1	12/2019	Jang
2018/0183089	A1	6/2018	Xu et al.	2019/0393542	A1	12/2019	Jang
2018/0183107	A1	6/2018	Zhamu et al.	2019/0393543	A1	12/2019	Zhamu et al.
2018/0190983	A1	7/2018	Fukui et al.	2020/0006770	A1	1/2020	Kamata et al.
2018/0205068	A1	7/2018	Takeuchi et al.	2020/0014057	A1	1/2020	Xu et al.
2018/0205080	A1	7/2018	Toya et al.	2020/0020895	A1	1/2020	Collins et al.
2018/0212241	A1	7/2018	Liu et al.	2020/0028147	A1	1/2020	Zhamu et al.
2018/0219048	A1	8/2018	Seo et al.	2020/0028158	A1	1/2020	Zhamu et al.
2018/0241080	A1	8/2018	Jena et al.	2020/0028205	A1	1/2020	Zhamu et al.
2018/0248185	A1	8/2018	Johnson et al.	2020/0044250	A1	2/2020	Toya et al.
2018/0248189	A1	8/2018	Pan et al.	2020/0052042	A1	2/2020	Seo et al.
2018/0248190	A1	8/2018	Pan et al.	2020/0052325	A1	2/2020	Zhamu et al.
2018/0254481	A2	9/2018	Kamata et al.	2020/0052350	A1	2/2020	Zhamu et al.
2018/0277894	A1	9/2018	Pan et al.	2020/0057085	A1	2/2020	Wasson et al.
2018/0277913	A1	9/2018	Pan et al.	2020/0058936	A1	2/2020	Horikawa et al.
2018/0287150	A1	10/2018	Goodenough et al.	2020/0067077	A1	2/2020	Pan et al.
2018/0294474	A1	10/2018	Zhamu et al.	2020/0067079	A1	2/2020	Pan et al.
2018/0294476	A1	10/2018	Zhamu et al.	2020/0067080	A1	2/2020	Pan et al.
2018/0301707	A1	10/2018	Pan et al.	2020/0067101	A1	2/2020	Pan et al.
2018/0307113	A1	10/2018	Bass et al.	2020/0083543	A1	3/2020	Choi et al.
2018/0323474	A1	11/2018	Liu et al.	2020/0091507	A1	3/2020	Zhamu et al.
2018/0342731	A1	11/2018	Ravet et al.	2020/0098997	A1	3/2020	Watabe et al.
2018/0342733	A1	11/2018	Goodenough et al.	2020/0144595	A1	5/2020	Lin
2018/0342737	A1	11/2018	Zhamu et al.	2020/0144600	A1	5/2020	Zhong et al.
2018/0347069	A1	12/2018	Toya et al.	2020/0144605	A1	5/2020	Su et al.
2018/0351200	A1	12/2018	Zhamu et al.	2020/0153068	A1	5/2020	Licht
2018/0351201	A1	12/2018	Zhamu et al.	2020/0207881	A1	7/2020	Ohkimoto et al.
2018/0375156	A1	12/2018	Zhamu et al.	2020/0230596	A1	7/2020	Holmes et al.
2019/0006671	A1	1/2019	Uekusa et al.	2020/0235421	A1	7/2020	Zhou et al.
2019/0020025	A1	1/2019	Matsumoto et al.	2020/0243838	A1	7/2020	Jang
2019/0027542	A1	1/2019	Watabe et al.	2020/0243854	A1	7/2020	Jang
2019/0027788	A1	1/2019	Liu et al.	2020/0253047	A1	8/2020	Lai et al.
2019/0044137	A1	2/2019	Zhamu et al.	2020/0259105	A1	8/2020	Watabe et al.
2019/0051905	A1	2/2019	Zhamu et al.	2020/0259200	A1	8/2020	Nie et al.
2019/0067732	A1	2/2019	Zhamu et al.	2020/0266426	A1	8/2020	Zhamu et al.
2019/0071319	A1	3/2019	Rodriguez et al.	2020/0280012	A1	9/2020	Mitsumori et al.
2019/0088882	A1	3/2019	Seo et al.	2020/0280054	A1	9/2020	Jang
2019/0097210	A1	3/2019	Chen et al.	2020/0280055	A1	9/2020	Jang
2019/0115591	A1	4/2019	Zhamu et al.	2020/0287206	A1	9/2020	Jang
2019/0148715	A1	5/2019	Liu et al.	2020/0287207	A1	9/2020	Jang
2019/0165365	A1	5/2019	Zhamu et al.	2020/0295267	A1	9/2020	Kanamoto et al.
2019/0165374	A1	5/2019	Zhamu et al.	2020/0295355	A1	9/2020	Kanno et al.
2019/0173079	A1	6/2019	Zhamu et al.	2020/0295399	A1	9/2020	Zhou et al.
2019/0173082	A1	6/2019	Zhamu et al.	2020/0313174	A1	10/2020	Wang et al.
2019/0180951	A1	6/2019	Yokota et al.	2020/0321614	A1	10/2020	Banerjee et al.
2019/0189936	A1	6/2019	Watabe et al.	2020/0326356	A1	10/2020	Wasson et al.
2019/0190022	A1	6/2019	Kim et al.	2020/0350373	A1	11/2020	Ohsawa et al.
2019/0207124	A1	7/2019	Kanamoto et al.	2020/0350503	A1	11/2020	Seo et al.
2019/0207200	A1	7/2019	Zhamu et al.	2020/0350508	A1	11/2020	Seo et al.
2019/0259583	A1	8/2019	Baba et al.	2020/0350589	A1	11/2020	Jang
2019/0273257	A1	9/2019	Siu et al.	2020/0354220	A1	11/2020	Pedersen et al.
2019/0305290	A1	10/2019	Singh et al.	2020/0358090	A1	11/2020	Jang
2019/0312272	A1	10/2019	Goodenough et al.	2020/0358141	A1	11/2020	Jang
				2020/0365902	A1	11/2020	Jang
				2020/0366180	A1	11/2020	Mills
				2020/0381728	A1	12/2020	Uekusa et al.
				2020/0395566	A1	12/2020	Ishisone et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2020/0403555	A1	12/2020	Mills
2021/0021001	A1	1/2021	Chang et al.
2021/0021003	A1	1/2021	Chen et al.
2021/0028451	A1	1/2021	Johnson et al.
2021/0028507	A1	1/2021	Su et al.
2021/0028509	A1	1/2021	Su et al.
2021/0043929	A1	2/2021	Huang et al.

OTHER PUBLICATIONS

Zima, Vítězslav, et al. "Ion-exchange properties of alkali-metal redox-intercalated vanadyl phosphate." *Journal of Solid State Chemistry* 163.1 (2002): 281-285.

Lii, Kwang-Hwa, and Wei-Chuan Liu. "RbVOPO₄ and CsVOPO₄, Two Vanadyl (IV) Orthophosphates with an Intersecting Tunnel Structure and Discrete VO₅ Pyramids." *Journal of Solid State Chemistry* 103.1 (1993): 38-44.

Yakubovich, O. V., O. V. Karimova, and O. K. Mel'nikov. "The mixed anionic framework in the structure of Na₂ {MnF [PO₄]}. " *Acta Crystallographica Section C: Crystal Structure Communications* 53.4 (1997): 395-397.

Schindler, M., F. C. Hawthorne, and W. H. Baur. "Crystal chemical aspects of vanadium: polyhedral geometries, characteristic bond valences, and polymerization of (VO_n) polyhedra." *Chemistry of Materials* 12.5 (2000): 1248-1259.

Panin, Rodion V., et al. "Crystal Structure, Polymorphism, and Properties of the New Vanadyl Phosphate Na₄VO (PO₄)₂." *Chemistry of materials* 16.6 (2004): 1048-1055.

Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "The Structure Determination of a New Mixed Mono-Arsenate K₂V₂O₂ (AsO₄)₂." 3ème Conférence Internationale sur le Soudage, le CND et l'Industrie des Matériaux et Alliages (IC-WNDT-MI'12). Centre de Recherche Scientifique et Technique en Soudage et Contrôle (CSC), 2012.

Glasser, Leslie, and C. Richard A. Catlow. "Modelling phase changes in the potassium titanyl phosphate system." *Journal of Materials Chemistry* 7.12 (1997): 2537-2542.

Fedotov, Stanislav S., et al. "AVPO₄F (A=Li, K): A 4 V Cathode Material for High-Power Rechargeable Batteries." *Chemistry of Materials* 28.2 (2016): 411-415.

Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "K₂V₂O₂ (AsO₄)₂." *Acta Crystallographica Section E: Structure Reports Online* 68.7 (2012): i54-i54.

Yakubovich, O. V., V. V. Kireev, and O. K. Mel'nikov. "Refinement of crystal structure of a Ge-analogue of natisite Na₂ {TiGeO₄} and prediction of new phases with anionic {MTO₅} radicals." *Crystallography Reports* 45.4 (2000): 578-584.

Boudin, S., et al. "Review on vanadium phosphates with mono and divalent metallic cations: syntheses, structural relationships and classification, properties." *International Journal of Inorganic Materials* 2.6 (2000): 561-579.

Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO₄/ α -LiVOPO₄ Cathodes." *Electrochemical and Solid-State Letters* 3, No. 10 (2000): 460-462.

Quackenbush, Nicholas F., Linda Wangoh, Bohua Wen, Ruibo Zhang, Youngmin Chung, Natasha Chernova, Zehua Chen et al. "Interfacial Effects of Electrochemical Lithiation of Epsilon-VOPO₄ and Evolution of the Electronic Structure." In *Meeting Abstracts*, No. 6, pp. 491-491. The Electrochemical Society, 2015.

Whittingham, M. Stanley. "Lithium batteries and cathode materials." *Chemical reviews* 104, No. 10 (2004): 4271-4302.

Winter, Martin, and Ralph J. Brodd. "What are batteries, fuel cells, and supercapacitors ?." *Chemical reviews* 104, No. 10 (2004): 4245-4270.

Melot, Brent C., and J-M. Tarascon. "Design and preparation of materials for advanced electrochemical storage." *Accounts of chemical research* 46, No. 5 (2013): 1226-1238.

Whittingham, M. Stanley. "Ultimate limits to intercalation reactions for lithium batteries." *Chemical reviews* 114, No. 23 (2014): 11414-11443.

Zaghib, K., A. Mauger, F. Gendron, and C. M. Julien. "Surface effects on the physical and electrochemical properties of thin LiFePO₄ particles." *Chemistry of Materials* 20, No. 2 (2008): 462-469.

Liu, Hao, Fiona C. Strobridge, Olaf J. Borkiewicz, Kamila M. Wiaderek, Karena W. Chapman, Peter J. Chupas, and Clare P. Grey. "Capturing metastable structures during high-rate cycling of LiFePO₄ nanoparticle electrodes." *Science* 344, No. 6191 (2014).

Hautier, Geoffroy, Anubhav Jain, Shyue Ping Ong, Byoungwoo Kang, Charles Moore, Robert Doe, and Gerbrand Ceder. "Phosphates as lithium-ion battery cathodes: an evaluation based on high-throughput ab initio calculations." *Chemistry of Materials* 23, No. 15 (2011): 3495-3508.

Siu, Carrie, Ieuan D. Seymour, Sylvia Britto, Hanlei Zhang, Jatinkumar Rana, Jun Feng, Fredrick O. Omenya et al. "Enabling multi-electron reaction of ϵ -VOPO₄ to reach theoretical capacity for lithium-ion batteries." *Chemical communications* 54, No. 56 (2018): 7802-7805.

Lim, S. C., J. T. Vaughey, W. T. A. Harrison, L. L. Dussack, A. J. Jacobson, and J. W. Johnson. "Redox transformations of simple vanadium phosphates: the synthesis of ϵ -VOPO₄." *Solid State Ionics* 84, No. 3-4 (1996): 219-226.

Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO₄ / α -LiVOPO₄ Cathodes." *Electrochemical and Solid State Letters* 3, No. 10 (2000): 460.

Song, Yanning, Peter Y. Zavalij, and M. Stanley Whittingham. " ϵ -VOPO₄: electrochemical synthesis and enhanced cathode behavior." *Journal of the Electrochemical Society* 152, No. 4 (2005): A721.

Lin, Yuh-Chieh, Bohua Wen, Kamila M. Wiaderek, Shawn Sallis, Hao Liu, Saul H. Lapidus, Olaf J. Borkiewicz et al. "Thermodynamics, kinetics and structural evolution of ϵ -LiVOPO₄ over multiple lithium intercalation." *Chemistry of Materials* 28, No. 6 (2016): 1794-1805.

Chen, Zehua, Qiyuan Chen, Liquan Chen, Ruibo Zhang, Hui Zhou, Natasha A. Chernova, and M. Stanley Whittingham. "Electrochemical Behavior of Nanostructured ϵ -VOPO₄ over Two Redox Plateaus." *Journal of The Electrochemical Society* 160, No. 10 (2013): A1777.

Azmi, Bustam M., Hasanah S. Munirah, Tatsumi Ishihara, and Yusaku Takita. "Optimized LiVOPO₄ for cathodes in Li-ion rechargeable batteries." *Ionics* 11, No. 5-6 (2005): 402-405.

Bianchini, Mateos, J. M. Ateba-Mba, Philippe Dagault, Elena Bogdan, D. Carlier, Emmanuelle Suard, Christian Masquelier, and Laurence Croguennec. "Multiple phases in the ϵ -VPO₄O—LiVPO₄O—Li₂VPO₄O system: a combined solid state electrochemistry and diffraction structural study." *Journal of Materials Chemistry A* 2, No. 26 (2014): 10182-10192.

Harrison, Katharine L., Craig A. Bridges, Carlo U. Segre, C. Daniel Varnado Jr, Danielle Applestone, Christopher W. Bielawski, Mariapan Parans Paranthaman, and Arumugam Manthiram. "Chemical and electrochemical lithiation of LiVOPO₄ cathodes for lithium-ion batteries." *Chemistry of Materials* 26, No. 12 (2014): 3849-3861.

Chernova, Natasha A., Megan Roppolo, Anne C. Dillon, and M. Stanley Whittingham. "Layered vanadium and molybdenum oxides: batteries and electrochromics." *Journal of Materials Chemistry* 19, No. 17 (2009): 2526-2552.

* cited by examiner

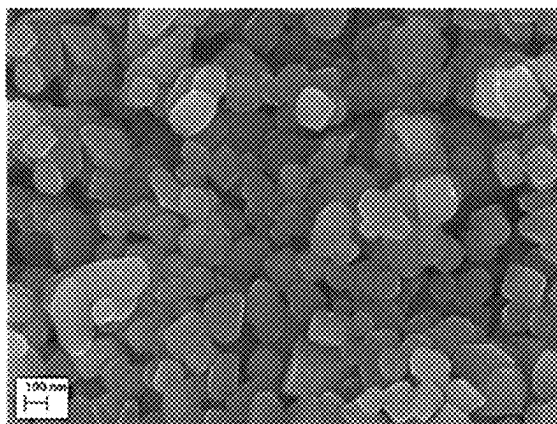


Fig. 1A
Morphological and structure
characterization of ϵ -VOPO₄.
SEM image

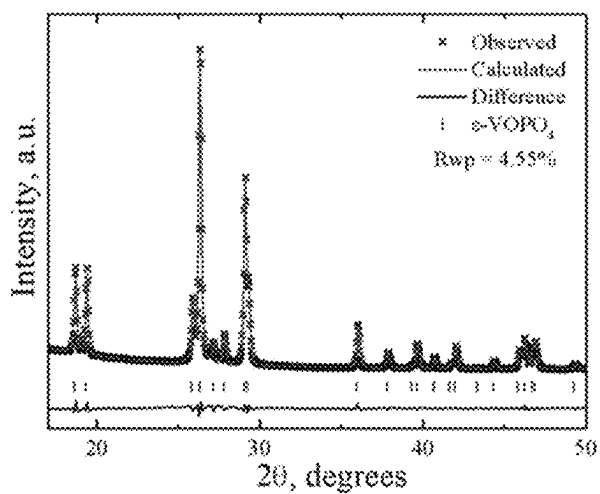


Fig. 1B
XRD pattern with Rietveld
refinement of the as synthesized
 ϵ -VOPO₄.

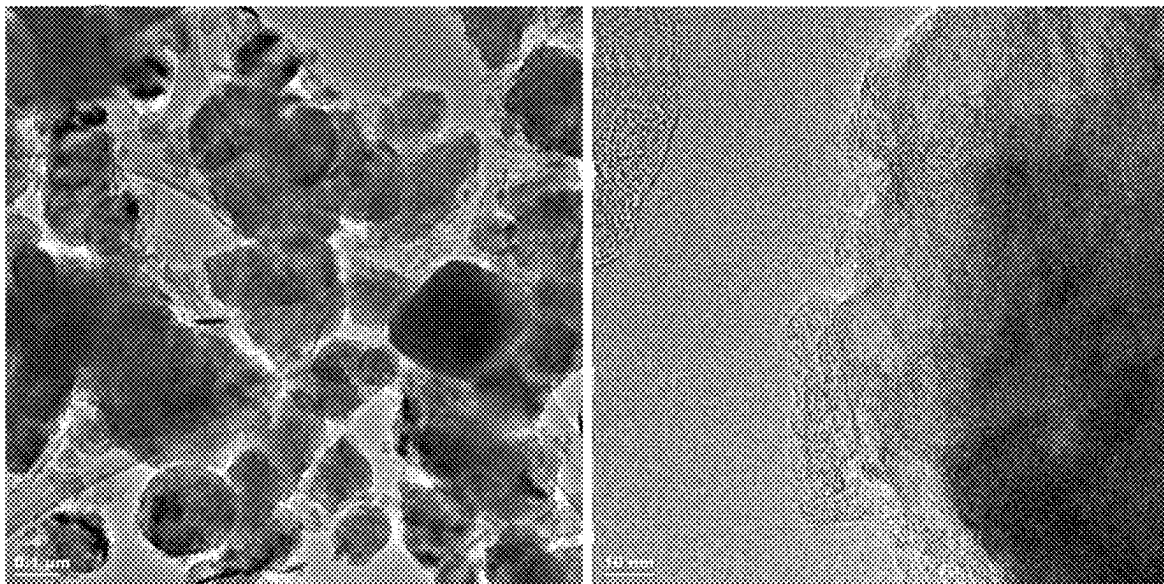


Fig. 2
TEM images of ϵ -VOPO₄ hand ground with graphene nanoplatelets for
electrode preparation

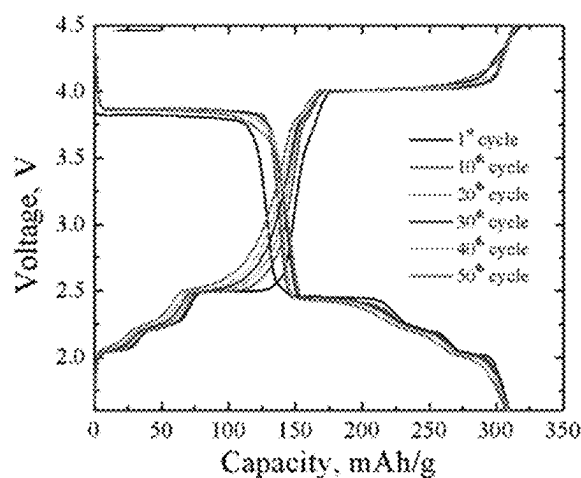


Fig. 3A

Galvanostatic charge-discharge curves of ϵ -VOPO₄ from 1.6 to 4.5V

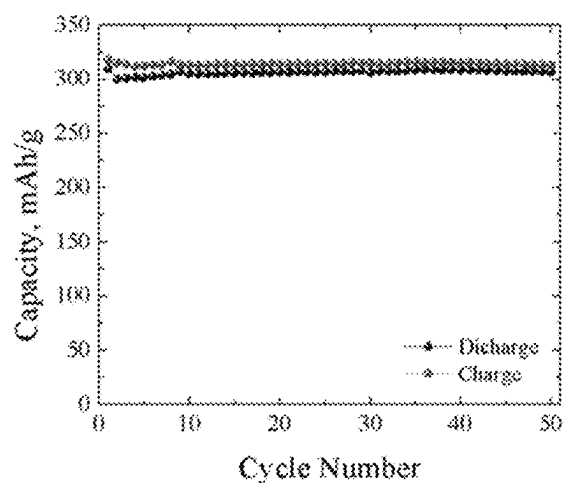


Fig. 3B

cycle performance at C/50, 1C=2 Li

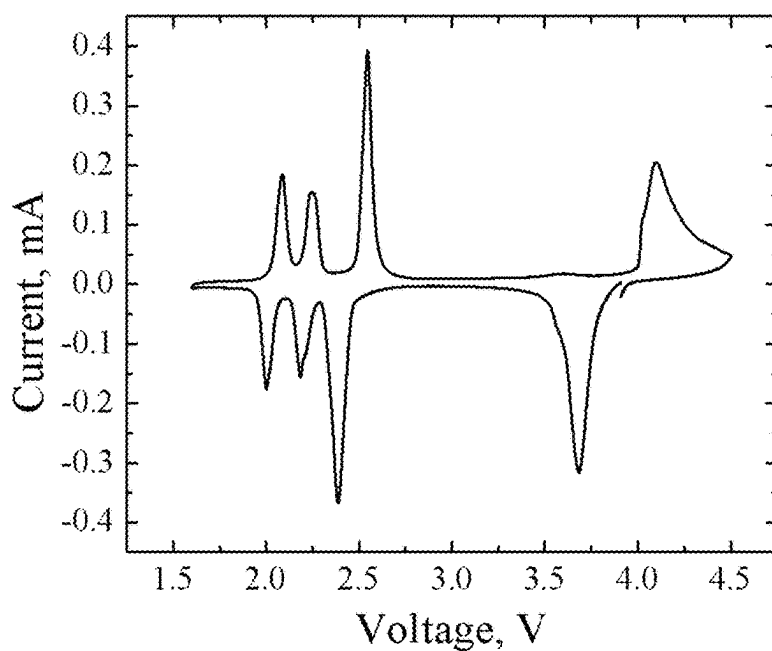


Fig. 4

CV curve profile of ϵ -VOPO₄ at a scan rate of 0.02mV/s

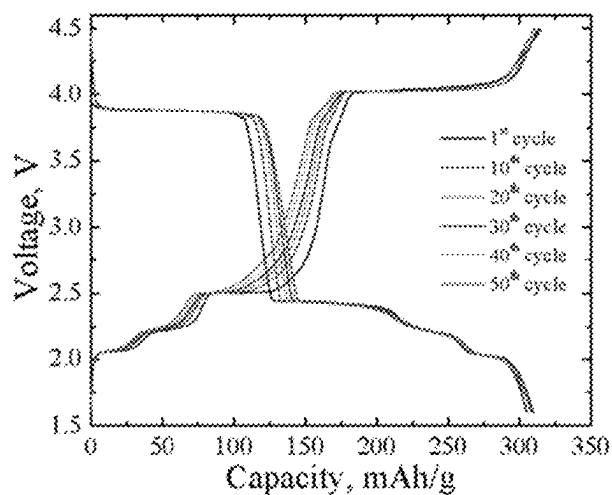


Fig. 5A

Galvanostatic charge-discharge curves of ϵ -VOPO₄ from 1.6 to 4.5V

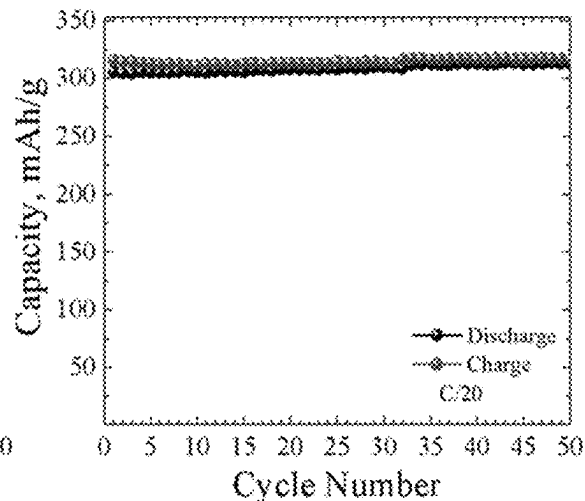


Fig. 5B

Cycle performance at C/20, 1C=2Li

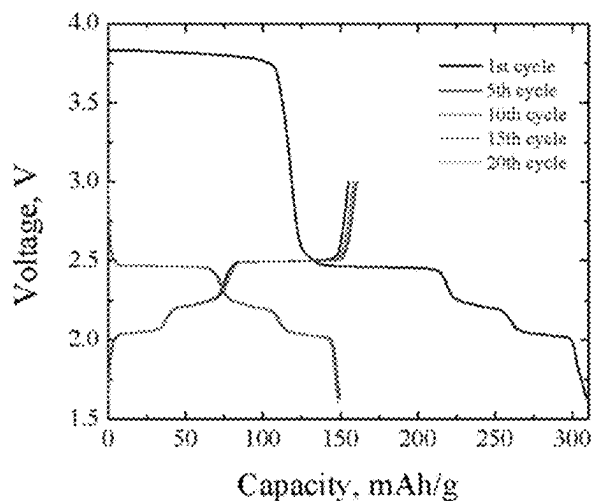


Fig. 6A

Galvanostatic charge-discharge Curves of ϵ -VOPO₄ at the low voltage region, from 1.6 to 3.0 V

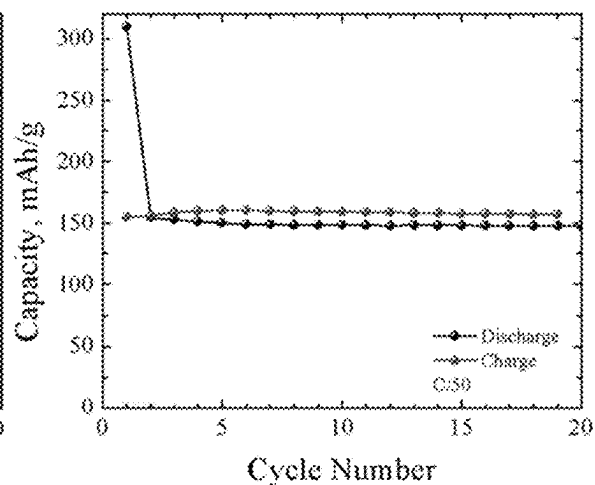


Fig. 6B

Cycle performance at C/50, 1C=2Li

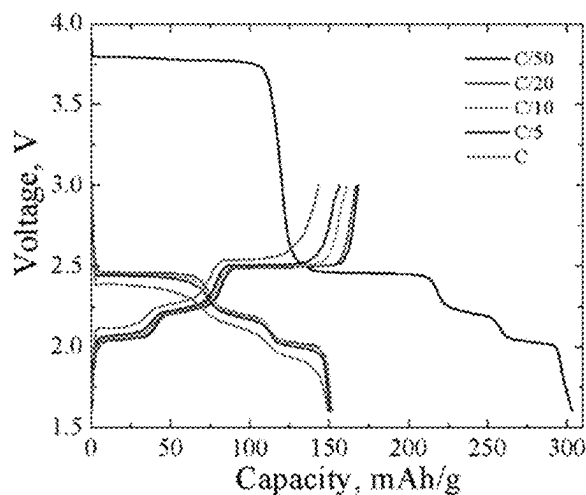


Fig. 7A

Cycling curves of ϵ -VOPO₄ in the low voltage region, from 1.6 - 3.0 V, at different rate

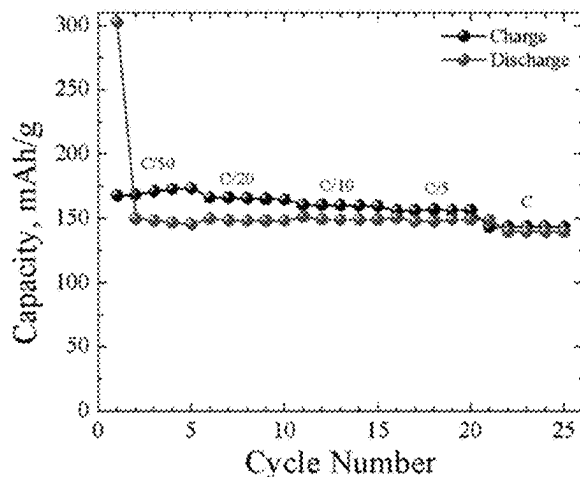


Fig. 7B

Cycle performance

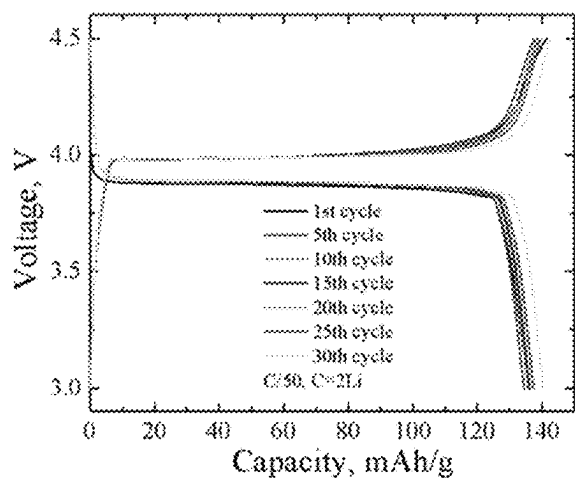


Fig. 8A

Galvanostatic charge-discharge curves of ϵ -VOPO₄ at the high voltage region, from 3.0 to 4.5V

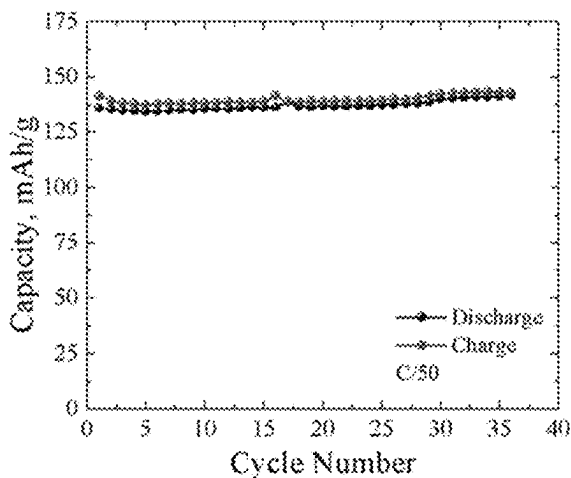


Fig. 8B

Cycle performance at C/50, 1C=2Li

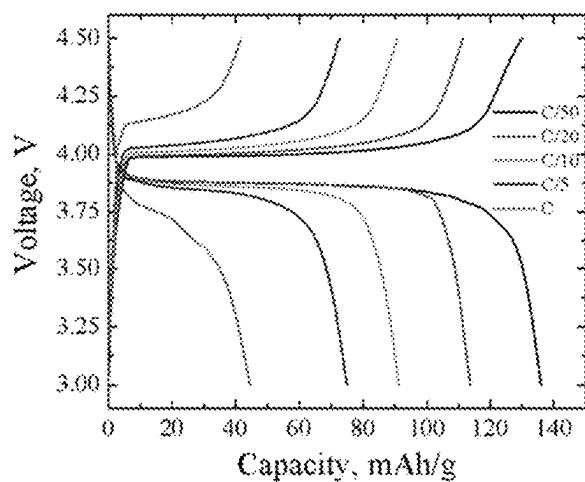


Fig. 9A

Cycle curves of ϵ -VOPO₄ at high voltage region, from 3.0 to 4.5 V, at different current rates

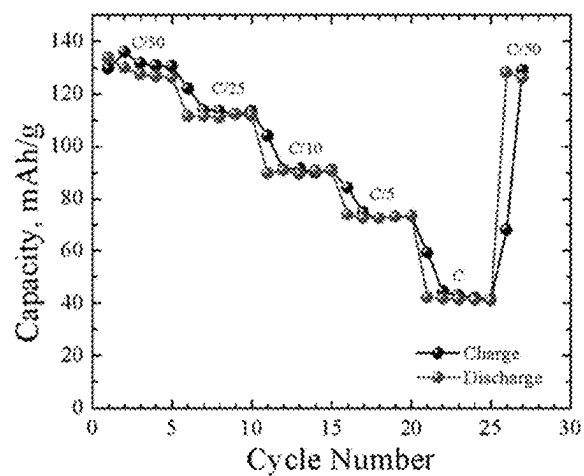


Fig. 9B

Cycle performance

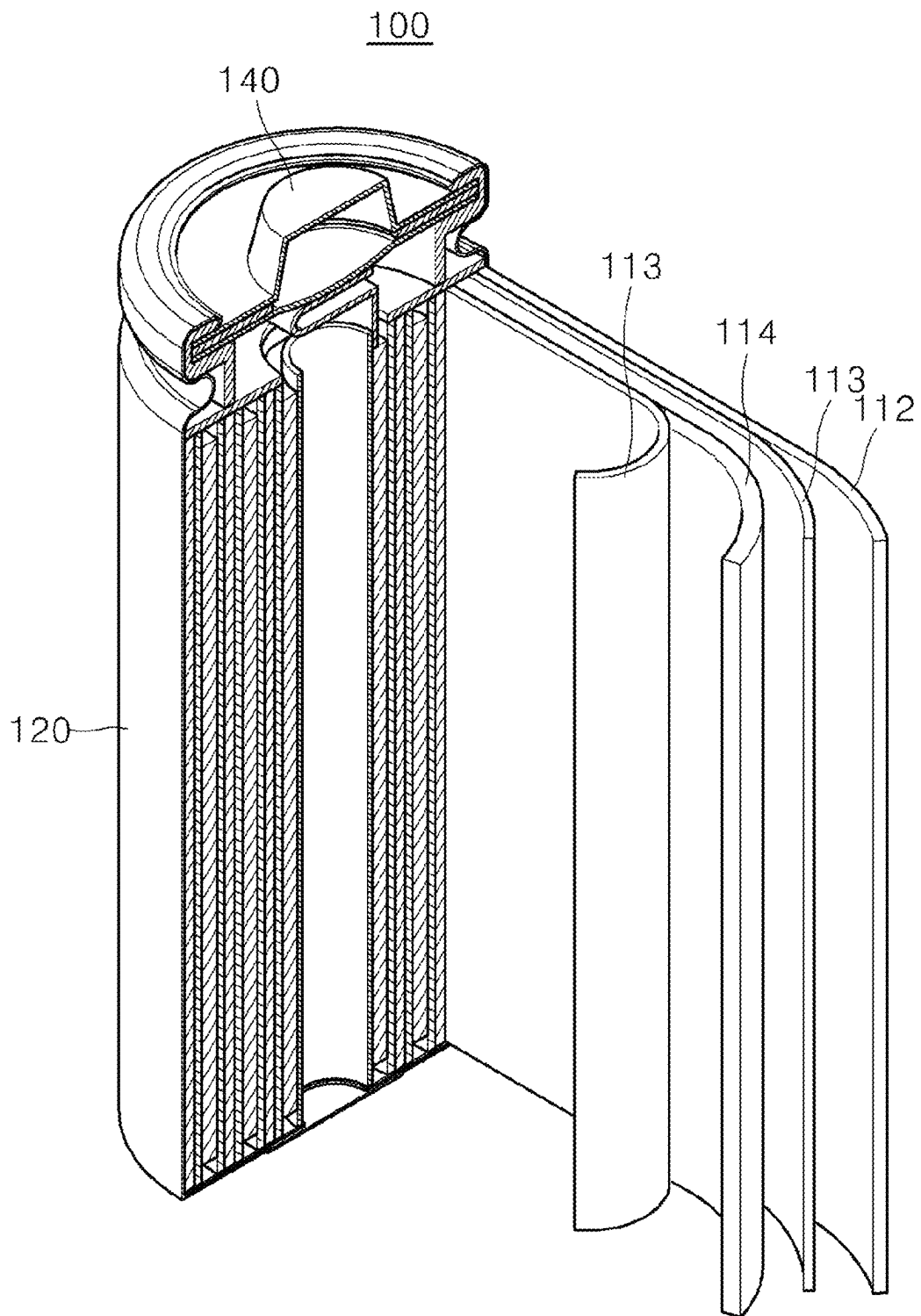


Fig. 10

Prior Art

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 ϵ -VOPO₄ CATHODE FOR LITHIUM ION BATTERIES**CROSS REFERENCE TO RELATED APPLICATIONS**

The present invention is a Continuation of U.S. patent application Ser. No. 18/354,493, filed Jul. 18, 2023, now U.S. Pat. No. 12,029,957, issued Jun. 4, 2024, which is a Continuation of U.S. patent application Ser. No. 17/670,470, filed Feb. 13, 2022, now U.S. Pat. No. 11,715,929, issued Aug. 1, 2023, which is a Division of U.S. patent application Ser. No. 16/291,617, filed Mar. 4, 2019, now U.S. Pat. No. 11,251,430, issued Feb. 15, 2022, which is a Non-Provisional of and claims benefit of priority from, U.S. Patent Application No. 62/638,893, filed Mar. 5, 2018, the entirety of which is expressly incorporated herein by reference.

STATEMENT OF FEDERALLY FUNDED RESEARCH

This invention was made with government support under contract DE-SC0012583 awarded by the Department of Energy. The government has certain rights in the invention.

FIELD OF THE INVENTION

The present invention relates to the field of electrode materials for batteries, and more particularly to an ϵ -vanadyl phosphate cathode having near theoretical dual lithium ion capacity.

BACKGROUND OF THE INVENTION

Energy conversion and storage have become more and more important in transportation, commercial, industrial, residential, and consumer applications. In particular, large-scale implementation of renewable energy, increasing ubiquity of portable electronics, and the next generation of electric vehicles require inexpensive and efficient energy storage systems.

A lithium secondary battery includes a cathode, an anode, and a separator. The separator may be a solid electrolyte, or an additional element with a liquid electrolyte. During discharging of the lithium secondary battery, oxidation reaction occurs in the anode due to deintercalation of lithium ions, while reduction reaction occurs in the cathode due to intercalation of lithium ions. The vice versa processes take place during the battery charging. The electrolyte has selective conductivity only ions, and thus transfers lithium ions between the cathode and the anode. Lithium ions intercalated into an electrode of a battery lead to charge neutrality with electrons entered into the electrode, and thus serve as media storing electric energy in the electrode. Accordingly, the quantity of electric energy storable by the battery is dependent upon the quantity of lithium ions intercalated into the electrode. Although basic performance of the lithium secondary battery, such as operating voltage and energy density, is dependent upon the materials of the cathode and anode, the electrolyte also needs to have high-ion conductivity, electrochemical stability and thermal stability to ensure high performance of the lithium secondary battery.

A typical lithium-ion battery electrolyte consists of a lithium salt and a solvent. Because of the high operating voltage, the solvent is typically anhydrous, with organic solvents now common, e.g., glyme. Phosphazenes and phosphoranimines have been proposed as an alternate non-

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flammable electrolyte. Other nitrogen, sulfur, phosphorus, silicon, compounds are also known as electrolyte additives or electrolytes. The electrolyte needs to be electrochemically stable in a voltage range where reduction and oxidation proceed in the anode and cathode, respectively.

As the use of lithium secondary batteries is expanding to electric vehicles and power storage fields, electrode active materials for use at high voltages emerged and became available. Use of a relatively low-potential anode active material and a relatively high-potential cathode active material has led to a narrower potential window of the electrolyte, so that the electrolyte is more likely to decompose on a surface of the cathode/anode. Lithium secondary batteries for electric vehicles and power storage are likely to be exposed to external high-temperature environment conditions, and the temperatures of these lithium secondary batteries may rise during instantaneous charging and discharging. Accordingly, lifetime and stored energy quantity of the lithium secondary battery may be reduced in such high-temperature environment conditions.

Typical lithium secondary battery technologies are discussed in U.S. Pat. No. 9,819,054, expressly incorporated herein by reference its entirety.

The non-aqueous solvent, which is in the electrolyte of a lithium secondary battery according to the above-described embodiments, may serve as a migration medium of ions involved in electrochemical reactions of the battery. Any suitable non-aqueous solvent that is commonly used in the art may be used. For example, the non-aqueous solvent may be an organic carbonate compound, an ester compound, an ether compound, a ketone compound, an alcohol compound, an aprotic bipolar solvent, or a combination thereof. The carbonate compound may be an open chain carbonate compound, a cyclic carbonate compound, a fluoro carbonate derivative thereof, or a combination thereof.

The electrolyte useful for the battery is one which does not chemically react with the anode or with the cathode during storage, and permits the migration of ions to intercalate the cathode-active material and vice-versa (during the discharge and charging cycles, respectively). The electrolyte may be present in a pure state (in the form of a solid, fused solid or liquid) or it may be conveniently dissolved in a suitable solvent. As a general rule, the electrolyte material should consist of a compound of the same species as that which is selected for the anode-active material. Thus, useful electrolytes may be conveniently represented by the general formula LY wherein L is a cationic moiety selected from the same materials useful as the anode-active material and Y is an anionic moiety or moieties such as halides, sulfates, nitrates, beta-aluminas, phosphofluorides, perchlorates and rubidium halide. The electrolyte may be present in a pure state in the form of a solid, fused solid (i.e. molten salt) or liquid or it may be conveniently dissolved in a suitable solvent which does not generally hydrolyze or degrade under conditions within the battery. Such electrolytes include ketones, esters, ethers, organic carbonates (such as propylene carbonate), organic lactones, organic nitriles, nitrohydrocarbons, organic sulfoxides, etc. and mixtures thereof. Where the solvent is utilized, the electrolyte salt may be present in a concentration determined by the desired solution conductivity, solubility and chemical reactivity. The electrolyte may include additives to reduce flammability, such as phosphazenes, e.g., cyclic phosphazenes.

Non-limiting examples of the chain carbonate compound are diethyl carbonate ("DEC"), dimethyl carbonate, ("DMC"), dipropyl carbonate ("DPC"), methylpropyl carbonate ("MPC"), ethylpropylcarbonate ("EPC"), methyl-

ethyl carbonate ("MEC"), and a combination thereof. Non-limiting examples of the cyclic carbonate compound are ethylene carbonate ("EC"), propylenecarbonate ("PC"), butylene carbonate ("BC"), fluoroethylene carbonate ("FEC"), vinylene carbonate ("VEC"), and a combination thereof. Non-limiting examples of the fluorocarbonate compound are fluoroethylene carbonate ("FEC"), 4,5-difluoroethylene carbonate, 4,4-difluoroethylene carbonate, 4,4,5-trifluoroethylene carbonate, 4,4,5,5-tetrafluoroethylene carbonate, 4-fluoro-5-methylethylene carbonate, 4-fluoro-4-methylethylene carbonate, 4,5-difluoro-4-methylethylene carbonate, 4,4,5-trifluoro-5-methylethylene carbonate, trifluoromethylethylene carbonate, and a combination thereof. The carbonate compound may include a combination of cyclic carbonate and chain carbonate, in consideration of dielectric constant and viscosity of the electrolyte. For example, an amount of a cyclic carbonate compound may be at least 10% by volume based on a total volume of the non-aqueous organic solvent. The carbonate compound may be a mixture of such chain carbonate and/or cyclic carbonate compounds as described above with a fluorocarbonate compound. The fluorocarbonate compound may increase solubility of a lithium salt to improve ionic conductivity of the electrolyte, and may facilitate formation of the thin film on the anode. In some embodiments, the fluorocarbonate compound may be fluoroethylene carbonate ("FEC"). An amount of the fluorocarbonate compound may be from about 1 to about 30 percent by volume ("volume %") based on a total volume of the non-aqueous organic solvent. When the amount of the fluorocarbonate compound is within this range, the electrolyte may have an appropriate viscosity to provide desired effects thereof.

Non-limiting examples of the ester compound are methyl acetate, ethyl acetate, n-propyl acetate, dimethyl acetate, methyl propionate ("MP"), ethyl propionate, γ -butyrolactone, decanolide, valerolactone, mevalonolactone, caprolactone, and methyl formate. Non-limiting examples of the ether compound are dibutyl ether, tetraglyme, diglyme, 1,2-dimethoxyethane, 1,2-diethoxyethane, ethoxymethoxyethane, 2-methyltetrahydrofuran, and tetrahydrofuran. An example of the ketone compound is cyclohexanone. Non-limiting examples of the alcohol compound are ethyl alcohol and isopropyl alcohol.

Examples of the aprotic solvent are nitriles (such as R—CN, wherein R is a C₂–C₂₀ linear, branched, or cyclic hydrocarbon-based moiety that may include a double-bond, an aromatic ring or an ether bond), amides (such as formamide and dimethylformamide), dioxolanes (such as 1,2-dioxolane and 1,3-dioxolane), methylsulfoxide, sulfolanes (such as sulfolane and methylsulfolane), 1,3-dimethyl-2-imidazolidinone, N-methyl-2-pyrrolidinone, nitromethane, trimethyl phosphate, triethyl phosphate, trioctyl phosphate, and triester phosphate.

The non-aqueous organic solvent may be used alone or in a combination of at least two solvents. In the latter case, a mixing ratio of the at least two non-aqueous organic solvents may be appropriately adjusted depending on a desired performance of the battery.

The non-aqueous organic solvent may further include an aromatic hydrocarbon organic solvent in the carbonate solvent. The carbonate solvent and the aromatic hydrocarbon organic solvent may be mixed, for example, in a volume ratio of about 1:1 to about 30:1. Examples of the aromatic hydrocarbon organic solvent are benzene, fluorobenzene, 1,2-difluorobenzene, 1,3-difluorobenzene, 1,4-difluorobenzene, 1,2,3-trifluorobenzene, 1,2,4-trifluorobenzene, chlorobenzene, 1,2-dichlorobenzene, 1,3-dichlorobenzene, 1,4-

dichlorobenzene, 1,2,3-trichlorobenzene, 1,2,4-trichlorobenzene, iodobenzene, 1,2-diiodobenzene, 1,3-diiodobenzene, 1,4-diiodobenzene, 1,2,3-triiodobenzene, 1,2,4-triiodobenzene, 2-fluorotoluene, 3-fluorotoluene, 4-fluorotoluene, 2,3-difluorotoluene, 2,4-difluorotoluene, 2,5-difluorotoluene, 2,6-difluorotoluene, 3,4-difluorotoluene, 3,5-difluorotoluene, 2,3,4-trifluorotoluene, 2,3,5-trifluorotoluene, 2,3,6-trifluorotoluene, 3,4,5-trifluorotoluene, 2,4,5-trifluorotoluene, 2,4,6-trifluorotoluene, 2-chlorotoluene, 3-chlorotoluene, 4-chlorotoluene, 2,3-dichlorotoluene, 2,4-dichlorotoluene, 2,5-dichlorotoluene, 2,6-dichlorotoluene, 2,3,4-trichlorotoluene, 2,3,5-trichlorotoluene, 2,3,6-trichlorotoluene, 3,4,5-trichlorotoluene, 2,4,5-trichlorotoluene, 2,4,6-trichlorotoluene, 2-iodotoluene, 3-iodotoluene, 4-iodotoluene, 2,3-diiodotoluene, 2,4-diiodotoluene, 2,5-diiodotoluene, 2,6-diiodotoluene, 3,4-diiodotoluene, 3,5-diiodotoluene, 2,3,4-triiodotoluene, 2,3,5-triiodotoluene, 2,3,6-triiodotoluene, 3,4,5-triiodotoluene, 2,4,5-triiodotoluene, 2,4,6-triiodotoluene, o-xylene, m-xylene, p-xylene, and combinations thereof.

Suitable electrolyte salts include, for example, LiPF₆, LiBF₄, LiClO₄, LiTFSI. Suitable solvents may be any solvent which is stable within the electrochemical window of the cell and is inert to other components of the cell. Examples of suitable solvents include carbonate solvents such as ethylene carbonate, diethyl carbonate, and propylene carbonate, organic ethers such as dioxolane, dimethyl ether and tetrahydrofuran and organic nitriles such as acetonitrile.

Additionally, the electrolyte may be a nonaqueous polymer electrolyte such as a gel polymer electrolyte, a solid ceramic electrolyte. In one embodiment, the electrolyte may include additives such as fluoroethylene carbonate (FEC) in order to, for example, improve cycling.

The lithium salt may be any suitable lithium salt that is commonly used for lithium batteries. Examples of the lithium salt for the non-aqueous electrolyte are LiPF₆, LiBF₄, LiSbF₆, LiAsF₆, LiSbF₆, LiCF₃SO₃, Li(CF₃SO₂)₃C, Li(CF₃SO₂)₂N, LiC₄F₉SO₃, LiClO₄, LiAlO₄, LiAlCl₄, LiAlF₄, LiBPh₄, LiBioCl₁₀, CH₃SO₃Li, C₄F₉SO₃Li, (CF₃SO₂)₂NLi, LiN(C_xF_{2x+1}SO₂)(C_yF_{2y+1}SO₂) (wherein x and y are natural numbers), CF₃CO₂Li, LiCl, LiBr, LiI, LiBOB (lithium bisoxalato borate), lower aliphatic carboxylic acid lithium, lithium terphenylborate, lithium imide, and a combination thereof. These lithium salts may be used as a supporting electrolytic salt.

A concentration of the lithium salt may be within a range known to one of ordinary skill in the art. The concentration of the lithium salt is not specifically limited, and in some embodiments, may be in a range of about 0.1 molar ("M") to about 2.0 M in the electrolyte to improve practical performance of a lithium battery. When the concentration of the lithium salt is within this range, the electrolyte may have appropriate conductivity and appropriate viscosity for improved performance, and may improve mobility of lithium ions.

In some embodiments, the electrolyte for a lithium battery may further include an additive to facilitate the formation of a stable solid electrolyte interphase ("SEI") or a thin film on a surface of an electrode to provide improved cycle characteristics.

Non-limiting examples of the additive are tris(trimethylsilyl)phosphate ("TMSPa"), lithium difluoro oxalate borate ("LiFOB"), vinylene carbonate ("VC"), propane sulfone ("PS"), succinonitrile ("SN"), LiBF₄, a silane compound having a functional group able to form a siloxane bond (for example, acryl, amino, epoxy, methoxy, ethoxy, or vinyl),

and a silazane compound such as hexamethyldisilazane. These additives may be used alone or in a combination of at least two thereof.

An amount of the additive may be from about 0.01 wt % to about 10 wt % based on a total weight of the non-aqueous organic solvent. For example, the amount of the additive may be from 0.05 wt % to about 10 wt %, in some embodiments, from about 0.1 wt % to about 5 wt %, and in some other embodiments, from about 0.5 wt % to about 4 wt %, based on the total weight of the non-aqueous organic solvent. However, the amount of the additive is not particularly limited unless the additive significantly hinders improvement in capacity retention rate of a lithium battery including the electrolyte. The lithium battery may be manufactured using a method known in the art.

According to the above-embodiments of the present disclosure, the lithium secondary battery may have a thin film formed on the surface of the cathode due to oxidation of at least a part of the additive in the electrolyte during initial charging of the lithium secondary battery. Thus, the lithium secondary battery may have improved capacity retention characteristics, lifetime characteristics and high-rate characteristics even when charged at a high operating voltage of about 4.0 V to about 5.5 V, for example, a voltage about 4.3 V to about 5.5 V. An additive in the electrolyte may enhance formation of a thin film on a surface of the cathode, the thin film having a thickness of, for example, about 0.05 nanometers ("nm") to about 100 nm. For example, the thin film may have a thickness of about 0.1 nm to about 80 nm, and in some embodiments, about 0.5 nm to about 50 nm. The thin film on the cathode surface may effectively prevent oxidation of the electrolyte on the cathode surface so that the conduction of lithium ions is not impeded.

FIG. 10 is an exploded perspective view of a lithium secondary battery 100 according to an embodiment. Although the lithium secondary battery 100 illustrated in FIG. 10 is cylindrical, embodiments of the present disclosure are not limited thereto, and lithium secondary batteries according to embodiments may be of a rectangular type or a pouch type. Lithium secondary batteries may be classified as lithium-ion batteries, lithium-ion polymer batteries, or lithium polymer batteries, according to the type of separator and/or electrolyte included therein. In addition, lithium batteries may be classified as cylindrical type, rectangular type, coin type, or pouch type, according to the shape thereof. Lithium batteries may also be classified as either bulk type or thin film type, according to the size thereof. Lithium secondary batteries according to embodiments may have any appropriate shape. The structure of a lithium secondary battery and a method of manufacturing the same are known in the art, so a detailed description thereof will not be recited here. Referring to FIG. 10, the lithium secondary battery 100, which is cylindrical, includes an anode 112, a cathode 114, a separator 113 disposed between the anode 112 and the cathode 114, and an electrolyte (not shown) impregnated into the anode 112, the cathode 114 and the separator 113, a battery case 120, and a sealing member 140 sealing the battery case 120. The lithium secondary battery 100 is manufactured by sequentially stacking the anode 112, the cathode 114, and the separator 113 upon one another to form a stack, rolling the stack in a spiral form, and accommodating the rolled up stack in the battery case 120. The cathode 114 includes a cathode current collector, and a cathode active material layer disposed on the cathode current collector. The cathode current collector may have a thickness of about 3 μm to about 500 μm . The cathode current collector is not particularly limited, and may be

formed of any material so long as it has a suitable conductivity without causing chemical changes in the fabricated battery. Examples of the cathode current collector include copper, stainless steel, aluminum, nickel, titanium, sintered carbon, copper or stainless steel that is surface-treated with carbon, nickel, titanium or silver, and aluminum-cadmium alloys. In addition, the cathode current collector may be processed to have fine irregularities on surfaces thereof so as to enhance adhesive strength of the cathode current collector to the cathode active material, and may be used in any of various forms including films, sheets, foils, nets, porous structures, foams, and non-woven fabrics.

Known cathode active material layers include a cathode active material, a binder, and a conducting agent.

Known cathode active materials include lithium-containing metal oxides, e.g., at least one of a composite oxide of lithium with a metal selected from Co, Mn, Ni, and a combination thereof.

During initial cycling, a solid electrolyte interphase layer (SEI layer) forms in an electrolyte battery, representing insoluble breakdown products of the electrolyte in combination with other battery components, such as electrode material. The SEI layer serves to protect the electrolyte from further free radical reactions during overvoltage periods, e.g., during charging.

The compounds listed above as cathode active materials may have a surface coating layer (hereinafter, "coating layer"). Alternatively, a mixture of a compound without a coating layer and a compound having a coating layer, the compounds being selected from the compounds listed above, may be used. The coating layer may include at least one compound of a coating element selected from oxide, hydroxide, oxyhydroxide, oxycarbonate, and hydroxycarbonate of the coating element. The compounds for the coating layer may be amorphous or crystalline. The coating element for the coating layer may be magnesium (Mg), aluminum (Al), cobalt (Co), potassium (K), sodium (Na), calcium (Ca), silicon (Si), titanium (Ti), vanadium (V), tin (Sn), germanium (Ge), gallium (Ga), boron (B), arsenic (As), zirconium (Zr), or mixtures thereof. The coating layer may be formed using any method that does not adversely affect the physical properties of the cathode active material when a compound of the coating element is used. For example, the coating layer may be formed using a spray coating method, a dipping method, or any other method known to one of ordinary skill in the art.

The binder strongly binds positive cathode active material particles together and to a current collector. Examples of the binder are, but not limited to, polyvinyl alcohol, carboxymethyl cellulose, hydroxypropyl cellulose, diacetyl cellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, a polymer including ethylene oxide, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene-butadiene rubber ("SBR"), acrylated SBR, epoxy resin, and nylon.

Known electrodes include a conducting agent used to provide conductivity to electrodes. Any electron conducting material that does not induce chemical change in batteries may be used. Examples of the conducting agent include natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, carbon fibers, and metallic materials, including copper, nickel, aluminum, and silver, in powder or fiber form. The conducting agent may include a single conductive material, such as a polyphenylene derivative, or a combination of at least two conductive materials.

The amounts of the cathode active material, the binder, and the conducting agent may be equivalent to those commonly used in lithium batteries. For example, a weight ratio of the cathode active material to a mixture of the conducting agent and the binder may be from about 98:2 to about 92:8, and in some embodiments from about 95:5 to about 90:10. A mixing ratio of the conducting agent to the binder may be, but not limited, from about 1:1.5 to about 1:3. The known cathode active materials may have, for example, an operating voltage range of about 4.0 V to ≥ 5.5 V.

An exemplary lithium secondary battery **100**, shown in FIG. **10**, is cylindrical, and includes an anode **112**, a cathode **114**, a separator **113** disposed between the anode **112** and the cathode **114**, and an electrolyte (not shown) impregnated into the anode **112**, the cathode **114** and the separator **113**, a battery case **120**, and a sealing member **140** sealing the battery case **120**. The lithium secondary battery **100** is manufactured by sequentially stacking the anode **112**, the cathode **114**, and the separator **113** upon one another to form a stack, rolling the stack in a spiral form, and accommodating the rolled up stack in the battery case **120**.

The cathode **114** includes a cathode current collector, and a cathode active material layer disposed on the cathode current collector. The cathode current collector may have a thickness of about 3 μm to about 500 μm . The cathode current collector is not particularly limited, and may be formed of any material so long as it has a suitable conductivity without causing chemical changes in the fabricated battery. Examples of the cathode current collector include copper, stainless steel, aluminum, nickel, titanium, sintered carbon, copper or stainless steel that is surface-treated with carbon, nickel, titanium or silver, and aluminum-cadmium alloys. In addition, the cathode current collector may be processed to have fine irregularities on surfaces thereof so as to enhance adhesive strength of the cathode current collector to the cathode active material, and may be used in any of various forms including films, sheets, foils, nets, porous structures, foams, and non-woven fabrics. The cathode active material layer includes a cathode active material, a binder, and a conducting agent.

The anode active layer includes an anode active material, a binder, and optionally a conducting agent. The anode active material is not particularly limited, and may be selected from any anode active materials used in the art. Non-limiting examples of the anode active material are lithium metal, a lithium metal alloy, a transition metal oxide, a doped or undoped lithium material, and a material that allows reversible intercalation and deintercalation of lithium ions, which may be used as a mixture or in combination of at least two thereof. The lithium metal alloy may be an alloy of lithium with a metal selected from sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), francium (Fr), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), silicon (Si), antimony (Sb), lead (Pb), indium (In), zinc (Zn), barium (Ba), radium (Ra), germanium (Ge), aluminum (Al), and tin (Sn). Non-limiting examples of the transition metal oxide are tungsten oxide, molybdenum oxide, titanium oxide, lithium titanium oxide, vanadium oxide, and lithium vanadium oxide. Examples of the material that allows doping or undoping of lithium therein are Si, Sn, Al, Ge, Pb, Bi, Sb, and a Si—Y alloy (where Y is an alkali metal, an alkali earth metal, a Group 11 element, a Group 12 element, a Group 13 element, a Group 14 element, a Group 15 element, a Group 16 element, a transition metal, a rare earth element, and a combination thereof, except for Sn. For example, Y may be magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), radium (Ra), scandium (Sc), yttrium (Y),

titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), seaborgium (Sg), technetium (Tc), rhenium (Re), iron (Fe), lead (Pb), ruthenium (Ru), osmium (Os), rhodium (Rh), iridium (Ir), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), zinc (Zn), cadmium (Cd), boron (B), aluminum (Al), gallium (Ga), tin (Sn), indium (In), titanium (Ti), germanium (Ge), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), sulfur (S), selenium (Se), tellurium (Te), or combinations thereof.

The material that allows reversible intercalation and deintercalation of lithium ions may be any carbonaceous anode active material that is commonly used in a lithium battery. Examples of such carbonaceous materials are crystalline carbon, amorphous carbon, or mixtures thereof. Non-limiting examples of the crystalline carbon are natural graphite, artificial graphite, expanded graphite, graphene, fullerene soot, carbon nanotubes, and carbon fiber. Non-limiting examples of the amorphous carbon are soft carbon (carbon sintered at low temperatures), hard carbon, mesophase pitch carbides, and sintered corks. The carbonaceous anode active material may be in, for example, spherical, planar, fibrous, tubular, or powder form.

The binder strongly binds anode active material particles together and to the anode current collector. Non-limiting examples of the binder are polyvinyl alcohol, carboxymethyl cellulose, hydroxypropyl cellulose, diacetyl cellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, a polymer including ethylene oxide, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene-butadiene rubber ("SBR"), acrylated SBR, epoxy resin, and nylon.

The conducting agent is used to provide conductivity to the anode. Any electron conducting material that does not induce chemical change in batteries may be used. Examples of the conducting agent are carbonaceous materials, such as natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, carbon fibers, and the like; metal-based materials, such as copper (Cu), nickel (Ni), aluminum (Al), silver (Ag), and the like, in powder or fiber form; and conductive materials, including conductive polymers, such as a polyphenylene derivative, and mixtures thereof.

The amounts of the anode active material, the binder, and the conducting agent may be equivalent to those commonly used in lithium batteries. For example, a weight ratio of the anode active material to a mixture of the conducting agent and the binder may be from about 98:2 to about 92:8. A mixing ratio of the conducting agent to the binder may be, but not limited to, from about 1:1.5 to about 1:3. The anode **112** and the cathode **114** may be each manufactured by mixing an active material, a conducting agent, and a binder in a solvent to prepare an active material composition, and coating the active material composition on a current collector. Any method of manufacturing such electrodes which is known to one of ordinary skill in the art may be used. Thus, a detailed description thereof will not be provided herein. Non-limiting examples of the solvent are N-methylpyrrolidone ("NMP"), acetone, and water.

A separator may be disposed between the cathode and the anode, according to the type of the lithium secondary battery. The separator helps maintain electrical isolation between the cathode and the anode. The separator may be any separator that is commonly used for lithium batteries. In particular, the separator may have low resistance to migration of ions in an electrolyte and have high electrolyte-retaining ability. Examples of the separator are glass fiber,

polyester, Teflon, polyethylene, polypropylene, polyvinylidene fluoride ("PVDF"), polytetrafluoroethylene ("PTFE"), and a combination thereof, each of which may be a nonwoven fabric or a woven fabric. The separator may be a single layer or a multi-layer. Examples of the separator are a polyethylene/polypropylene double-layer separator, polyethylene/polypropylene/polyethylene triple-layer separator, and a polypropylene/polyethylene/polypropylene triple-layer separator. The separator may have a pore diameter of about 0.01 to about 10 μm and a thickness of about 3 to about 100 μm . The electrolyte may be injected between the cathode 114 and the anode 112 with the separator 113 therebetween.

A separator may include fibers, particles, web, porous sheets, or other forms of material configured to reduce the risk of physical contact and/or short circuit between the electrodes. The separator may be a unitary element, or may include a plurality of discrete spacer elements such as particles or fibers. In some examples, the electrolyte layer may include a separator infused with an electrolyte solution. In some examples such as a polymer electrolyte, the separator may be omitted.

See: U.S. Pat. Nos. 8,318,353; 7,901,810; 7,790,319; 7,771,628; 7,759,008; 7,041,239; 6,872,492; 4,355,162; 4,376,709; 4,376,709; 4,459,406; 5,849,434; 6,844,047; 6,872,492; 6,872,492; 6,888,663; 6,958,860; 7,041,239; 7,338,734; 7,383,083; 7,759,008; 7,759,008; 7,771,628; 7,790,319; 7,901,810; 7,939,201; 7,955,733; 7,960,058; 7,964,308; 7,972,728; 7,988,746; 7,998,617; 8,003,250; 8,057,936; 8,067,117; 8,088,512; 8,148,009; 8,148,013; 8,158,090; 8,168,326; 8,206,468; 8,206,469; 8,218,351; 8,241,789; 8,277,975; 8,282,691; 8,318,353; 8,323,831; 8,323,832; 8,361,661; 8,420,258; 8,435,678; 8,445,135; 8,449,980; 8,481,208; 8,492,031; 8,580,430; 8,586,238; 8,617,430; 8,617,745; 8,652,683; 8,658,310; 8,709,647; 8,722,227; 8,722,227; 8,722,242; 8,734,539; 8,785,043; 8,821,763; 8,852,807; 8,932,762; 8,936,871; 8,951,676; 8,951,680; 8,993,171; 8,999,571; 9,065,093; 9,077,032; 9,293,790; 9,299,966; 9,305,716; 9,349,544; 9,362,562; 20020192553; 20020192553; 20020192553; 20040005265; 20040016632; 20040016632; 20040048157; 20040066556; 20040197654; 20040197654; 20040262571; 20040262571; 20050238961; 20050238961; 20050238961; 20060110580; 20060194112; 20060194112; 20060292444; 20070031732; 20070072034; 20070072034; 20070141468; 20070141468; 20070166617; 20070190418; 20070248520; 20070292747; 20070298317; 20080261113; 20080261113; 20090214944; 20090220838; 20090220838; 20090246636; 20090311597; 20100075225; 20100078591; 20100233545; 20100266899; 20100266899; 20100303701; 20110052473; 20110052986; 20110052986; 20110052986; 20110052995; 20110159381; 20110195306; 20110200848; 20110200848; 20110229765; 20110229765; 20110274948; 20120140378; 20120164499; 20120164499; 20120214071; 20120219856; 20120219859; 20120237828; 20120270109; 20120315538; 20130034780; 20130052492; 20130059211; 20130078523; 20130078524; 20130084474; 20130084474; 20130084474; 20130115521; 20130252112; 20130252114; 20130302697; 20130344367; 20130344367; 20140099557; 20140220450; 20140242445; 20140302403; 20140370388; 20150111105; 20150132650; 20150236342; 20150236349; 20150263382; 20150303474; 20150303474; 20150311565; 20150364753; 20160028114; 20160096334; 20160111720; and 20160164152, each of which is expressly incorporated herein by reference in its entirety.

See also:

- Berrah, Fadila, et al. "The vanadium monophosphates AVOPO_4 : Synthesis of a second form $\beta\text{-KVVOPO}_4$ and structural relationships in the series." *Solid state sciences* 3.4 (2001): 477-482.
- Zima, Vítězslav, et al. "Ion-exchange properties of alkali-metal redox-intercalated vanadyl phosphate." *Journal of Solid State Chemistry* 163.1 (2002): 281-285.
- Lii, Kwang-Hwa, and Wei-Chuan Liu. "RbVOPO₄ and CsVOPO₄, Two Vanadyl (IV) Orthophosphates with an Intersecting Tunnel Structure and Discrete VO₅ Pyramids." *Journal of Solid State Chemistry* 103.1 (1993): 38-44.
- Yakubovich, O. V., O. V. Karimova, and O. K. Mel'nikov. "The mixed anionic framework in the structure of $\text{Na}_2\{\text{MnF}[\text{PO}_4]\}$." *Acta Crystallographica Section C: Crystal Structure Communications* 53.4 (1997): 395-397.
- Schindler, M., F. C. Hawthorne, and W. H. Baur. "Crystal chemical aspects of vanadium: polyhedral geometries, characteristic bond valences, and polymerization of (VO n) polyhedra." *Chemistry of Materials* 12.5 (2000): 1248-1259.
- Panin, Rodion V., et al. "Crystal Structure, Polymorphism, and Properties of the New Vanadyl Phosphate $\text{Na}_4\text{VO}(\text{PO}_4)_2$." *Chemistry of materials* 16.6 (2004): 1048-1055.
- Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "The Structure Determination of a New Mixed Mono-Arsenate $\text{K}_2\text{V}_2\text{O}_7(\text{AsO}_4)_2$." *3ème Conférence Internationale sur le Soudage, le CND et l'Industrie des Matériaux et Alliages (IC-WNDT-MI'12)*. Centre de Recherche Scientifique et Technique en Soudage et Contrôle (CSC), 2012.
- Glasser, Leslie, and C. Richard A. Catlow. "Modelling phase changes in the potassium titanil phosphate system." *Journal of Materials Chemistry* 7.12 (1997): 2537-2542.
- Fedotov, Stanislav S., et al. "AVPO₄F (A=Li, K): A 4 V Cathode Material for High-Power Rechargeable Batteries." *Chemistry of Materials* 28.2 (2016): 411-415.
- Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "K₂V₂O₇(AsO₄)₂." *Acta Crystallographica Section E: Structure Reports Online* 68.7 (2012): i54-i54.
- Yakubovich, O. V., V. V. Kireev, and O. K. Mel'nikov. "Refinement of crystal structure of a Ge-analogue of natissite $\text{Na}_2\{\text{TiGeO}_4\}$ and prediction of new phases with anionic {MTOs} radicals." *Crystallography Reports* 45.4 (2000): 578-584.
- Boudin, S., et al. "Review on vanadium phosphates with mono and divalent metallic cations: syntheses, structural relationships and classification, properties." *International Journal of Inorganic Materials* 2.6 (2000): 561-579.

Each reference cited herein is expressly incorporated herein by reference in its entirety.

SUMMARY OF THE INVENTION

The present technology provides a vanadyl phosphates e-VOPO_4 cathode which has achieved multi-electron storage as lithium-ion battery cathode. Vanadyl phosphates in general have low intrinsic conductivity. A high efficiency battery cathode should have low electrical resistance. To overcome the conductivity problem, the cathode material is preferably nanosized, and coated with particles of a low activation energy conductive material, such as graphene or carbon nanotubes. This cathode utilizes the two redox

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couples of vanadium cation (i.e. V^{5+}/V^{4+} , V^{4+}/V^{3+}) to permit more than one lithium ion to be stored in the unit structure per vanadium ion. The involvement of the multiple redox processes of vanadium is reflected by the well separated high voltage plateau region at ~ 3.8 V and low voltage plateau region at ~ 2 V.

The two-electron redox property of vanadium results in a theoretical capacity of 305 mAh/g. In practical, maximum discharge capacity of over 300 mAh/g was obtained within the voltage region of 1.3-4.5 V vs. Li/Li^+ , which is over 90% of the theoretical value.

The electrode material is not limited to use in batteries, or as a cathode, or for use in lithium-ion electrolyte systems.

In addition to ϵ -VOPO₄ material, the cathode may further contain any cathode material suitable for lithium-ion insertion and release. Suitable auxiliary materials may include phosphate based materials such as FePO₄, VPO₄F, V₂(PO₄)₂F₃, FePO₄F, and V₂(PO₄)₃; oxides such as CoO₂, orthorhombic MnO₂, layered iron oxides FeO₂, chromium oxide CrO₂, layered Ni_{0.5}Mn_{0.5}O₂, and V₆O₁₅ nanorods; layer sulfides such as TiS₂; perovskite transition metal fluorides, or a mixture thereof.

The epsilon polymorph of vanadyl phosphate, ϵ -VOPO₄, made from the hydrothermally or more generally, solvothermally synthesized H₂VOPO₄, is a cathode material for lithium-ion batteries that has been optimized to reversibly intercalate two Li-ions to reach the full theoretical capacity at least 50 cycles with a coulombic efficiency of 98%. This material adopts a stable 3D tunnel structure and can extract two Li-ions per vanadium ion, giving a theoretical capacity of 305 mAh/g, with an upper charge/discharge plateau at around 4.0 V, and one lower at around 2.5 V. As ϵ -VOPO₄ is capable of reversibly intercalating more than one lithium ion into the structure, it stores and delivers more energy than current batteries in the market. Compared to LiFePO₄, ϵ -VOPO₄ has a higher electronic conductivity and higher energy density with the insertion of one Li-ion, 159 mAh/g at 4.0 V vs 170 mAh/g at 3.45 V. Overall, ϵ -VOPO₄ makes a great candidate for next generation of high energy Li-ion batteries. The nano-sized ϵ -VOPO₄ particles demonstrate enhanced electrochemistry and cyclability for potential applications in lithium-ion batteries.

Phosphate based materials have been considered as excellent cathode candidates because of their high stability and low cost. However, most phosphate cathodes show poor electronic conductivity and as a result, full capacity of the cathode can't be achieved in the traditional charge/discharge processes.

One approach to obtain a cathode of high capacity is to employ a transition metal capable of multiple electron transfer, and thus able to assume more than one lithium. Vanadium is well-known to be capable of transfer of two electrons, such as from the +5 to +3 oxidation state.

Vanadyl phosphate (VOPO₄) is a material combining the merits of vanadium and of phosphate and theoretically has the possibility to show high capacity as well as good stability as a cathode active material for a sodium battery. The vanadyl phosphates with formula of AVOPO₄ (A=alkali metal) form a class of materials which can serve as a multi-electron cathode. These cathodes can utilize the V^{3+} - V^{4+} - V^{5+} redox couples, during which two ions can be reversible stored in the structure instead of one. Therefore, this class of cathode materials is expected to exhibit much higher energy density than the traditional one-electron cathodes. By far, the two-electron behavior only has been observed in Li ion system (i.e. two-Li) in some different

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phases of VOPO₄ and LiVOPO₄, within a voltage window covering the $V^{3+} \rightarrow V^{5+}$ transition, which exhibits enhanced practical energy densities.

Vanadium phosphate materials have been described as cathode materials.

It is an object of this invention to a high energy density cathode active material for use in a lithium battery, using ϵ -VOPO₄ as an active material, and which comprises a conductivity enhancer comprising graphene or carbon nanotubes.

The cathode containing any of the above-listed materials may be mixed with other electrically conductive materials and binders. Examples of electrically conductive materials include carbon black and vapor ground carbon fibers. Examples of binders include polyvinylidene fluoride (PVDF), sodium alginate, and sodium carboxymethyl cellulose.

The cathode active material may be mixed with binders recognized by one of skill in the art as suitable for lithium-ion batteries. For example, suitable binders may include PVDF, polytetrafluoroethylene (PTFE), styrene butadiene rubber (SBR), and polyimide.

According to the present invention, the conductive additive is graphene or carbon nanotubes. The amount of conductive additive may be 1-10% by weight, and preferably about 5%.

The cathode active material may be present as a sheet, ribbon, particles, or other physical form. An electrode containing the cathode active material may be supported by a current collector. A current collector may include a metal or other electrically conducting material. The current collector may be formed of carbon, carbon paper, carbon cloth or a metal or noble metal mesh or foil.

U.S. Pat. No. 9,722,247 (Zhang et al.) discusses F-Vanadyl phosphates as high energy density cathode materials for rechargeable sodium batteries.

For example, U.S. Pat. No. 6,872,492 (Barker et al.) describes sodium ion batteries based on cathode materials of the general formula: $A_a M_b (XY_4)_c Z_d$. Example 4b describes synthesis of VOPO₄·H₂O and Examples 4c and 4d describe synthesis of NaVOPO₄. Charge and discharge of a cell containing a cathode of the NaVOPO₄ and a negative electrode of lithium metal is described. Sodium ion cells prepared are based on a carbon composite negative electrode and NaVOPO₄F as the positive electrode active material.

U.S. 2013/0034780 (Muldoon et al.) describes a magnesium battery and lists VOPO₄ as a suitable positive electrode active material.

U.S. 2004/0048157 (Neudecker et al.) describes a lithium solid state thin film battery containing a lithiated vanadium oxide film as an anode and as one possible cathode material, LiVOPO₄.

U.S. 2013/0260228 (Sano et al.) describes a lithium secondary battery having as a positive electrode material, a compound of the formula: $Li_a(M)_b(PO_4)_c F_d$. LiVOPO₄ is described in a preferred embodiment.

U.S. 2013/0115521 (Doe et al.) describes a magnesium secondary battery wherein the current collectors are coated with a thin protective coating. VOPO₄ is listed as a positive electrode active material.

U.S. 2012/0302697 (Wang et al.) describes a magnesium cell having a carbon or other graphitic material as a cathode active material. VOPO₄ is included in a list of other cathode active materials.

Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO₄/α-LiVOPO₄ Cathodes." *Electrochemical and Solid-State Letters* 3, no. 10

(2000): 460-462 discusses a ϵ -VOPO₄/α-LiVOPO₄ cathode for a lithium battery having a capacity of 100 mA/g.

Quackenbush, Nicholas F., Linda Wangoh, Bohua Wen, Ruibo Zhang, Youngmin Chung, Natasha Chernova, Zehua Chen et al. "Interfacial Effects of Electrochemical Lithiation of Epsilon-VOPO₄ and Evolution of the Electronic Structure." In Meeting Abstracts, no. 6, pp. 491-491. The Electrochemical Society, 2015 discusses the use of ϵ -VOPO₄ as a cathode material in lithium batteries, which achieves dual cation exchange only on particle surfaces, and not in the bulk material.

Seven distinct VOPO₄ structures or phases are known. All of the reported structures contain VO₆ octahedra sharing vertices with PO₄ tetrahedra. The oxygen polyhedron of vanadium is irregular so that it is often considered as a VO₅ square pyramid with a very short apical vanadyl bond (V=O) and a much more remote sixth oxygen atom (V . . . O). These seven phases can be distinguished as:

α_I : has a lamellar structure with alternating antiparallel V=O bonds pointing inside the layers.

α_{II} : also has a lamellar structure with antiparallel V=O bonds pointing outside the layers.

γ : is an intermediate form between α_I and α_{II} with half parallel V=O bonds pointing inside, half outside the layers.

δ : has antiparallel V=O bonds pointing half inside, half outside the layers. The vanadyl chains point to different directions in the unit cell.

ω : shows disordered vanadyl chains in the [1 0 0] and [0 1 0] directions of the tetragonal cell.

β : All vanadyl chains are parallel and tilted to form zigzag O=V . . . O=V chains.

ϵ : The structure is a distorted form of β -phase and differs in terms of tilted O=V . . . O angle.

It is therefore an object to provide an intercalation-type electrode composition for a lithium ion battery cathode, having a capacity of at least 275 mAh/g, and may have a capacity of at least 280 mAh/g, at least 290 mAh/g, at least 300 mAh/g, or at least 305 mAh/g, for example. The intercalation electrode composition may comprise a transition metal having a two-electron redox property, having a discharge capacity of at least 75% of theoretical value, at least 80% of theoretical value, at least 85% of theoretical value, or at least 90% of theoretical value.

The lithium ion battery cathode may comprise a transition metal, which undergoes a change in oxidation state of at least two between a charged and discharged state.

The intercalation electrode composition preferably comprises VOPO₄, most preferably in the epsilon form, i.e., ϵ -VOPO₄. The intercalation electrode composition preferably has a conductivity enhancer comprising graphene or carbon nanotubes. The intercalation electrode composition may comprise ϵ -VOPO₄ and at least 2.5% by weight graphene, at least 3.0% by weight graphene, at least 3.5% by weight graphene, at least 4.0% by weight graphene, at least 5% by weight graphene, at least 6% by weight graphene, at least 7% by weight graphene, at least 8% by weight graphene, at least 9% by weight graphene, or at least 10% by weight graphene.

The intercalation electrode composition may comprise, for example, at least 75% by weight ϵ -VOPO₄, at least 5% by weight graphene nanoplatelets, and at least 5% by weight of a poly vinylidene fluoride (PVDF) binder. The intercalation electrode composition may comprise 85% by weight ϵ -VOPO₄, at least 5% by weight graphene nanoplatelets, and 10% by weight binder. The intercalation electrode composition may comprise 75% by weight ϵ -VOPO₄, 15% by

weight graphene nano platelets, and 10% by weight of a poly vinylidene fluoride (PVDF) binder.

The intercalation electrode composition may be provided as a cathode in a battery comprising a lithium-ion anode, an electrolyte adapted to operate at a battery potential of at least 4.5V, a separator, and a supporting lithium salt.

The intercalation electrode composition may have a first state in which at least 80 mol % of a transition metal element is oxidized in a first oxidation state, and a second state in which at least 80 mol % of a transition metal element is oxidized in a second oxidation state, the first state and the second state differing by two, and the at least 80 mol % of the transition metal element in the first state is associated with two lithium ions per transition metal element ion.

Another object provides a lithium-ion battery cathode composition, comprising ϵ -VOPO₄, electrically conductive graphene in a ratio of at least 3% by weight of the ϵ -VOPO₄, and a binder, on a current collector substrate.

A further object provides an intercalation electrode composition for a lithium-ion cathode, having a dual lithium-ion exchange characteristic, having a capacity of about 125 mAh/g at a voltage exceeding 3.7 V, and a capacity of about 260 mAh/g at a voltage exceeding 2.0 V.

A still further object provides an intercalation electrode composition for a lithium-ion cathode, having a dual lithium-ion exchange characteristic, having an energy capacity of at least 850 mWh/g. The energy capacity may be at least 860 mWh/g, at least 870 mWh/g, an energy capacity of at least 880 mWh/g, an energy capacity of at least 890 mWh/g, or an energy capacity of at least 900 mWh/g.

Another object provides an electrode composition for a lithium-ion cathode, comprising ϵ -VOPO₄ having a theoretical capacity of 305 mA/g and an observed capacity of at least 275 mAh/g. The observed capacity may be at least 280 mAh/g, at least 285 mAh/g, at least 290 mAh/g, at least 295 mAh/g, or at least 300 mAh/g.

A further object provides an electrode, comprising a ϵ -VOPO₄ lithium ion exchange active material, graphene nanoplatelets, and a binder, having an energy density of 900 mWh/g.

A still further object provides an electrode, comprising a ϵ -VOPO₄ lithium ion exchange active material, having a current-voltage profile which displays peaks at about 2.1 V, 2.25 V, 2.5 V, and 3.9 V, representing a dual-lithium-ion exchange per vanadium ion of at least 90%.

An object also provides a method of making a lithium-ion battery cathode, comprising: hydrothermally or solvothermally generating ϵ -VOPO₄; mixing the ϵ -VOPO₄ with graphene nanoplatelets and a binder to form a mixture; and coating a current collector with a slurry of the mixture. The graphene nanoplatelets may have a surface area of 750 m²/g, for example. The binder may be polyvinylidene fluoride. The weight ratio of ϵ -VOPO₄, graphene nanoplatelets, and polyvinylidene fluoride may be 75:15:10.

The invention comprises the cathode material, an electrode formed of the cathode material, and electrochemical devices, e.g., a secondary battery, formed using the electrode.

Other objects will be apparent from a review hereof.

BRIEF DESCRIPTION OF THE DRAWINGS

These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:

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FIG. 1A shows morphological and structure characterization of ϵ -VOPO₄ SEM image.

FIG. 1B shows an XRD pattern with Rietveld refinement of the as-synthesized ϵ -VOPO₄.

FIG. 2 shows TEM images of ϵ -VOPO₄ hand ground with graphene nanoplatelets for electrode preparation.

FIG. 3A shows galvanostatic charge-discharge curves of ϵ -VOPO₄ from 1.6 to 4.5 V at C/50.

FIG. 3B shows cycle performance of ϵ -VOPO₄ from 1.6 to 4.5 V at C/50, 1C=2 Li.

FIG. 4 shows a CV curve profile of ϵ -VOPO₄ at a scan rate of 0.02 mV/s.

FIG. 5A shows galvanostatic charge-discharge curves of ϵ -VOPO₄ from 1.6 to 4.5 V at C/20.

FIG. 5B shows cycle performance of ϵ -VOPO₄ from 1.6 to 4.5 V at C/20, 1C=2Li.

FIG. 6A shows galvanostatic charge-discharge curves of ϵ -VOPO₄ at the low voltage region, from 1.6 to 3.0 V.

FIG. 6B shows cycle performance in the low voltage region, 1.6 to 3.0 V, of ϵ -VOPO₄ at C/50, 1C=2Li.

FIG. 7A shows cycling curves of ϵ -VOPO₄ in the low voltage region, from 1.6-3.0 V, at different rates.

FIG. 7B shows rate test capacities of ϵ -VOPO₄ in the low voltage region, from 1.6 to 3.0 V.

FIG. 8A shows galvanostatic charge-discharge curves of ϵ -VOPO₄ at the high voltage region, from 3.0 to 4.5 V.

FIG. 8B shows cycle performance in the high voltage region, 3.0-4.5 V, of ϵ -VOPO₄ at C/50, 1C=2Li.

FIG. 9A shows Cycle curves of ϵ -VOPO₄ at high voltage region, from 3.0 to 4.5 V, at different current rates.

FIG. 9B shows rate test capacities of ϵ -VOPO₄ in the low voltage region, from 3.0 to 4.5 V.

FIG. 10 is an exploded perspective view of a lithium secondary battery according to an embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to the like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

It will be understood that when an element is referred to as being "on" another element, it can be directly in contact with the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present.

It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component,

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region, layer, or section without departing from the teachings of the present embodiments.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term "or" means "and/or." It will be further understood that the terms "comprises" and/or "comprising," or "includes" and/or "including" when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this general inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, are appropriate for use only if consistent with a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Thus, such resources as "Merriam Webster" (any version) are secondary to field of science-appropriate technical dictionaries and encyclopedias.

One or more embodiments will now be described in detail with reference to the following examples. However, these examples are not intended to limit the scope of the one or more embodiments.

The basis of the lithium-ion battery (LIB) uses lithium-ions to travel across the electrolyte and intercalate into the anode upon charge and into the cathode upon discharge [1]. Because they outperform competing primary batteries like lead-acid, alkaline, etc., as well as other rechargeable batteries such as nickel-metal hydride, nickel cadmium, etc., it is no wonder how the LIB has revolutionized and expanded the mobile electronics industry since 1991 [2]. While the performance and functionality of smartphones and laptops continue to improve, the development of LIBs need to catch up to match in terms of power and life cycle to expand into large energy storage applications.

Currently, the cathode material in the market is dominated by LiCoO₂. While it has an extremely high theoretical capacity of 274 mAh/g with an operating voltage around 3.6 V, the structure tends to undergo irreversible changes when more than 50% of the Li-ions are removed [3]. Environmental safety and the cost became major concerns as well, thus, drove the search for alternatives where the scarce and toxic cobalt is substituted by other metals such as in LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂, also known as NMC [4]. Extensive research for developing alternative cathode materials lead to study of metal phosphates, thus came the LiFePO₄. This olivine material has garnered a great deal of attention that it has been commercialized for portable and stationary systems by A123, BAE Systems and in China. In fact, Hydro-Quebec, MIT and A123 further improved this material with high-power performance that can charge and discharge within minutes through nanosizing and carbon coating [5]. Through a collaborative effort from NECCES-I, a fundamental study on this phenomenon determined a metastable reaction mechanism of LiFePO₄ which is why it can cycle at high rates [6].

Another way to leverage in the stability of phosphates and to increase energy storage is to incorporate a second electron. Hautier et al. plotted the mean voltage for each per

metal redox couple with respect to the capacity in a phosphate, giving way to consider two electron couples. Vanadium phosphate compounds were candidates within the acceptable voltage window [7]. The ϵ -VOPO₄ ↔ ϵ -LiVOPO₄ ↔ ϵ -Li₂VOPO₄ system has been regarded as one of the most promising and safe candidates to provide a two-electron reaction with a high theoretical capacity of 305 mAh/g and specific energy over 900 Wh/g [8, 15, 16]. This system has two redox potentials at useful potentials for storing energy, V³⁺ ↔ V⁴⁺ at 4.0 V and V⁴⁺ ↔ V⁵⁺ at 2.5 V. ϵ -VOPO₄ was first synthesized by Lim et al. by heating monoclinic H₂VOPO₄ in oxygen, and Kerr et al. measured the electrochemical reversibility at the high voltage plateau at 4.0 V [9, 10]. Previously, the synthesis and characterization of ϵ -VOPO₄ from two different phases of H₂VOPO₄ was reported, to discover that the electrochemical performance from the disordered tetragonal precursor was improved due to smaller particle size [11]. Fundamental studies were conducted on the structural evolution of ϵ -LiVOPO₄ and two intermediate phases in the low-voltage regime identified using DFT calculations backed up with X-ray pair distribution function analysis and X-ray absorption near edge structure measurements [12]. The insertion of two Li-ions into ϵ -VOPO₄ has been demonstrated, reaching the theoretical specific capacity of 305 mAh/g.

Example 1

Synthesis: ϵ -VOPO₄ was synthesized by calcining the monoclinic H₂VOPO₄ precursor, as reported by Song et al. [11] Stoichiometric amounts of VCl₃ (Sigma-Aldrich, 97%), and P₂O₅ (Sigma-Aldrich, ≥98%) were dissolved in 190 proof ethanol (Pharmco-AAPER). The solution was placed in a 4748 Type 125 mL PTFE-lined reactor (Parr Instrument Co.) and the reaction was set to 180° C. for 72 hours. The solvothermal product was collected by centrifugation and heated at 550° C. in flowing oxygen for 3 hours.

Materials Characterization: For XRD measurements, a Bruker D8 Advanced X-ray diffractometer equipped with Cu K α source, λ =1.54178 Å. The intensities were recorded within the 2 θ range from 10° to 80° with 2 θ steps of 0.02° from powder samples. The unit cell parameters were obtained by Rietveld refinement with the TOPAS program. Scanning electron microscopy (SEM) measurements were performed with a Zeiss Supra-55 VP field emission scanning electron, using both the secondary electron and InLens modes to determine the morphology and particle size, at an acceleration voltage of 10 kV. Transmission electron microscopy (TEM) imaging was performed using the FEI Titan 80-300 microscope with a field emission gun (FEG) and an image aberration corrector, operated at an acceleration voltage of 300 kV. The pristine material was dispersed on a copper grid coated with a lacey carbon film for high-resolution transmission electron microscopy (HRTEM) observation.

Electrochemistry: The electrodes were prepared by mixing the active material, ϵ -VOPO₄, with graphene nanoplatelets (surface area 750 m²/g, XG Sciences) as a carbon additive and polyvinylidene fluoride (PVDF, Aldrich) binder in a weight ratio of 75:15:10. The slurry was created by adding 1-methyl-2-pyrrolidinone (NMP, Aldrich) which was then laminated onto an aluminum foil 144 current collector and vacuum-dried overnight before use. The dried electrodes, of area 1.2 cm², contained 8-10 mg of active material and were assembled in 2325-type coin cells in a He-filled glovebox with a pure lithium chip (thickness 0.38 mm, Aldrich) as the counter and reference electrode. The elec-

trolyte used was lithium hexafluorophosphate (1 M LiPF₆) dissolved in ethylene carbonate (EC) and dimethyl carbonate (DMC) in 1:1 volume ratio with Celgard 2400 (Hoechst Celanese) as the separator. The electrochemical properties were investigated using the Bio-Logic VMP multichannel potentiostat. The cells were cycled galvanostatically in the high voltage region (3.0-4.5 V), low voltage region (1.6-3.0 V) and the whole voltage (1.6-4.5 V) window at C/50, where 1 C=2 Li or 305 mAh/g per gram of Li₂VOPO₄. Cells were also cycled at C/20 over the whole voltage range.

Hydrothermal or solvothermal synthesis has many unique advantages because it offers good control over the sample's purity and crystallinity, easy to scale up and low cost. This method can keep the overall particle size small and size distribution narrow which are vital features for good cathodic electrochemical performance. FIG. 1A shows the as-synthesized ϵ -VOPO₄ powder as nano-sized primary particles, ~100-200 nm, that are cuboid in shape. This material matched well with earlier reported results from Chen et al., where ϵ -VOPO₄ synthesized from monoclinic H₂VOPO₄ are made up of single crystals up to 200 nm [13]. Achieving small primary particles is important because it can improve the rate property for Li intercalation. Azmi et al. reported that smaller LiVOPO₄ particle size results in easier lithium-ion diffusion with enhanced columbic efficiency by improving the capacity of lithium deintercalation upon discharge and the decreasing the lithium intercalation potential upon charge [14]. It is also observed in FIG. 1A, that the nano-sized ϵ -VOPO₄ primary particles do not agglomerate nor form into secondary particles, providing a good surface area for the graphene or carbon nanotube additive to wrap around and assist in electron migration during the charge/discharge process.

The lack of primary particle agglomeration is attributed to the choice of solvent used for synthesis. By using 190 proof ethanol, ϵ -VOPO₄ results in loose particle morphology whereas 200 proof ethanol results in the formation of 2 μ m balls as secondary particles. By using this solvothermal synthesis route, the precursor was successfully synthesized and calcined to produce pure crystalline P-VOPO₄, as seen in FIG. 1B, resulting in sharp and narrow peaks in the x-ray diffraction pattern. The observed pattern matched very well to the calculated pattern with no impurities or other vanadyl phosphate phases, resulting in a low R_{wp} value of 4.55%.

Normally, ball-mill treatment is necessary to break up any agglomeration and secondary particles and to reduce the particle size for good electrochemical performance. However, since the as-synthesized ϵ -VOPO₄ is of nanometer size, there is no need to use this application, which helps preserve the crystal structure for better reversible intercalation chemistry. FIG. 2 shows HRTEM images of 75 wt. % ϵ -VOPO₄ that was hand milled with 15 wt. % graphene nanoplatelets in a mortar and pestle before adding 10 wt. % PDVF and NMP for electrode preparation. FIGS. 2A and 2B show graphene nanoplatelets forming a conductive network between every single ϵ -VOPO₄ primary particle. Upon closer inspection at FIG. 2B, HRTEM shows that the graphene nanoplatelets coated on the ϵ -VOPO₄ particle is around 10 nm thick.

Electrochemistry of ϵ -VOPO₄:

FIGS. 3A and 3B show ϵ -VOPO₄ cycled in the whole voltage window from 1.6 V to 4.5 V at C/50, capable of achieving a high discharge capacity of 305 mAh/g for at least 50 cycles. FIG. 2A displays the desired characteristic plateaus at ~4.0 V at the high voltage region and at ~2.5, 2.25, 2.0 V at the low voltage region. The drop from the high voltage region to the low voltage region is a step-like curve

and the hysteresis gap between the charge and discharge curve is very small. The high voltage region has a long plateau which extends the capacity to ~ 150 mAh/g, equivalent to ~ 1 Li. This corresponds to the redox potential of V^{3+} and V^{4+} where ϵ -VOPO₄ becomes ϵ -LiVOPO₄. The low voltage region has three plateaus at 2.5, 2.25 and 2.0 V which also extends the capacity to ~ 150 mAh/g, corresponding to the second intercalation of lithium where ϵ -LiVOPO₄ becomes ϵ -Li₂VOPO₄. The plateaus at the low voltage region has maintained step-like curves even after 35 cycles, suggesting good kinetics and the changes in the local structure may be reversible for easy Li intercalation.

Cyclic voltammetry (CV) curves was measured in the voltage window of 1.6 V to 4.5 V to understand the redox process of ϵ -VOPO₄ is shown in FIG. 4. There are four reduction peaks at certain voltages that correspond to four oxidation peaks at similar voltages. Each peak represents the reversible reaction between ϵ -VOPO₄ and ϵ -Li₂VOPO₄ that correspond to the voltage plateaus found upon galvanostatic charge and discharge cycling. Starting from the OCV point at 3.9 V, there is a single oxidation peak at 3.7 V that indicates electrochemical lithiation from ϵ -VOPO₄ to ϵ -LiVOPO₄. As the scan rate test moves to the low voltage region, there are three additional oxidation peaks. Each of the peaks signify the transition from ϵ -LiVOPO₄ to ϵ -Li₂VOPO₄ with intermediate stages in between. ϵ -LiVOPO₄ becomes ϵ -Li_{1.5}VOPO₄ at ~ 2.5 V, then it converts to ϵ -Li_{1.75}VOPO₄ at ~ 2.25 V and finally becomes ϵ -Li₂VOPO₄ at ~ 2.0 V. Reduction peaks appear as the voltage continues to sweep from the low to high voltage domain, indicating that the V^{5+} oxidation state of ϵ -VOPO₄ was recovered from V^{3+} of ϵ -Li₂VOPO₄. From ϵ -Li₂VOPO₄, it becomes ϵ -Li_{1.75}VOPO₄ at ~ 2.1 V, then ϵ -Li_{1.5}VOPO₄ at ~ 2.25 V and ϵ -LiVOPO₄ at ~ 2.5 V. No further reaction takes place until ~ 4.25 V where ϵ -LiVOPO₄ further reduces to become ϵ -VOPO₄.

FIGS. 5A and 5B show that even at a faster rate, ϵ -VOPO₄ can still deliver a discharge capacity of ~ 305 mAh/g for up to 40 cycles at C/20. The long high voltage plateau extending past 100 mAh/g is preserved and each of the characteristic steps in the low voltage region are clearly sustained with no signs of diminishing for up to 30 cycles. In FIG. 4A, the drop from the high voltage region to the low voltage region evolved to a slope-like curve, which helps make up for the shorter high voltage plateau in the beginning but might indicate a little hysteresis. In subsequent cycles, the high voltage plateau slightly increases. The capacity of the 1st high voltage discharge plateau was ~ 125 mAh/g and by the 35th cycle, it increased to ~ 150 mAh/g which is equivalent to 1 Li. The low voltage region seems to show the opposite trend. As the high voltage plateau starts to increase in capacity, the low voltage steps start to decrease as well to maintain the overall discharge capacity at ~ 305 mAh/g.

Cycling of ϵ -VOPO₄ has been separated at the high voltage region and the low voltage region to study the stability of the electrochemical curve and capacity without the influence of each other. FIGS. 6A and 6B show ϵ -VOPO₄ cycled in the low voltage region to study how long-term cycling affects the shape and length of these three distinct plateaus at 2.5 V, 2.25 V and 2.0 V which is in agreement with DFT calculations confirming the two intermediate phases at $x=1.5$ and 1.75 in the low-voltage regime [12]. The initial discharge curve in FIG. 6A is different because the cell was discharged from OCV first, delivering more than 300 mAh/g. Even after 30 cycles, FIG. 6A shows that each voltage step is clearly distinguished, delivering a reversible capacity of ~ 160 mAh/g, correlating to 1 Li. From then on,

the cell was continuously charged and discharged in the low voltage window, from 1.6 V to 3.0 V. FIG. 6B shows that the low voltage steps maintained ~ 160 mAh/g for up to 30 cycles with no sign of decay at all, suggesting good kinetics at the low voltage region. The charge-discharge profiles with steps agree with many reports, giving evidence to the existence of intermediate phases of ϵ -Li_xVOPO₄ where $x=1.5$ and 1.75 .

A rate test in the low voltage region was performed to study how faster cycling can affect the plateaus at 2.5 V, 2.25 V and 2.0 V, as shown in FIGS. 7A and 7B. To start the rate test at the low voltage region, the cell was first discharged from OCV to 1.6 V at C/50, delivering a discharge capacity of over 300 mAh/g. From C/50 to C/5, the low voltage plateaus still maintained a discharge capacity of ~ 150 mAh/g with clearly defined step-like features, as shown in FIG. 7A. When the rate increased to 1C, the discharge capacity is still ~ 150 mAh/g but the plateaus are more slope-in shape at slightly lower voltages. As the cycling rate increases, the difference between the charge and discharge capacities decreases, as shown in FIG. 7B. From C/50, the charge capacity is 175 mAh/g while the discharge is ~ 150 mAh/g. When the rate increased to 1C, the charge and discharge capacities are ~ 150 mAh/g, thereby increasing the coulombic efficiency to $\sim 100\%$. When cycled at the low voltage range, ϵ -VOPO₄ can reversibly intercalate one full lithium ion at the low voltage region, even at faster cycling rates.

FIGS. 8A and 8B show ϵ -VOPO₄ cycled in the high voltage region, from 3.0 V to 4.5 V, to study how long-term cycling affects the shape and capacity. In this high voltage window, there is a plateau at ~ 4.0 V that coincides with the $V^{3+/4+}$ redox where ϵ -VOPO₄ $\leftrightarrow$ LiVOPO₄. This high voltage plateau delivers a reversible capacity of ~ 140 mAh/g for up to 35 cycles which is close to 0.93 Li. This exceeds the previously reported results, where only 0.83 Li was inserted into ϵ -VOPO₄ and 0.65 Li was inserted into ϵ -LiVOPO₄. [11,12] After 30 cycles, the capacity slowly climbs to 150 mAh/g which corresponds to 1 Li which agrees with the trend in the galvanostatic charge-discharge curves in FIG. 4. Even after 30 cycles, FIG. 8B shows that ϵ -VOPO₄ delivers a reversible capacity of ~ 150 mAh/g, correlating to 1 Li. This plateau is step-like with no signs of fading after many cycles, suggesting easy reversible intercalation.

FIGS. 9A and 9B show how different rates can affect the high voltage plateau of ϵ -VOPO₄ at ~ 4.0 V. At C/50, the discharge capacity is around 130 mAh/g and the capacity decreases as the rate gets faster. By 1 C, the discharge capacity dropped to around 40 mAh/g. Despite the fast rate cycling of 1 C, the cell could deliver the high discharge capacity of 140 mAh/g after it was cycled back to C/50. This suggests that the structure was preserved, even at fast cycling, and can maintain a high discharge capacity when it was cycled back to C/50 from 1 C. It also seems that faster cycling leads to higher coulombic efficiency. It is evident that from C/50 to C, the coulombic efficiency increases. From C/25, some of the charge and discharge capacities are overlapping and by C/10, C/5 and C, the discharge capacities are practically the same as the charge. This means that at faster rates, it can de/intercalate lithium ion more efficiently.

CONCLUSIONS

The optimized morphology and nano particle size of ϵ -VOPO₄ is studied, as well as observations from transmission electron microscope to analyze the good carbon conductive network. By combining complementary character-

ization techniques of SEM, XRD and extensive electrochemical studies, the reversibility reaction of ϵ -VOPO₄ as a cathode material for lithium-ion batteries is elucidated.

Throughout this description all ranges described include all values and subranges therein, unless otherwise specified. Additionally, the indefinite article “a” or “an” carries the meaning of “one or more” throughout the description, unless otherwise specified. Where a numerical limit or range is stated, the endpoints are included. Also, all values and subranges within a numerical limit or range are specifically included as if explicitly written out.

The above description is provided to enable a person skilled in the art to make and use the invention, and is provided in the context of a particular application and its requirements. Various modifications to the preferred embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the invention. Thus, this invention is not intended to be limited to the embodiments shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein. In this regard, certain embodiments within the invention may not show every benefit of the invention, considered broadly.

REFERENCES

- M. S. Whittingham, Chem. Rev. 2014, 104, 4271-4301.
- M. Winter, R. J. Brodd, Chem. Reviews, 2004, 104, 4245-4270.
- B. C. Melot, J. M. Tarascon, Acc. Chem. Res., 2013, 46, 1226-1238.
- M. S. Whittingham, Chem. Rev. 2014, 114, 11414-11443.
- K. Zaghib, A. Mauger, F. Gendron, C. M. Julien, Chem. Mater., 2008, 20, 462-469.
- H. Liu, F. C. Strobridge, O. J. Borkiewicz, K. M. Wiaderek, K. W. Chapman, P. J. Chupas, C. P. Grey, Science, 2014, 344, 1252817.
- G. Hautier, A. Jain, S. P. Ong, B. Kang, C. Moore, R. Doe, G. Ceder, Chem. Mater. 2011, 23, 3495-3508.
- B. Wen, Q. Wang, Y. C. Lin, N. A. Chernova, K. Kharki, Y. Chung, F. Omenya, S. Sallis, L. F. J. Piper, S. P. Ong, M. S. Whittingham, Chem. Mater., 2016, 9, 3159-3170.
- S. C. Lim, J. T. Vaughey, W. T. A. Harrison, L. L. Dussack, A. J. Jacobson, J. W. Johnson, Solid State Ionics 1996, 84, 219-226.
- T. A. Kerr, Solid-State Lett. 1999, 3, 460.
- Song, Y.; Zavalij, P. Y.; Whittingham, M. S. J. Electrochem. Soc. 2005, 152, A721.
- Y. C. Lin, B. Wen, K. M. Wiaderek, S. Sallis, H. Liu, S. H. Lapidus, O. J. Borkiewicz, N. F. Quackenbush, N. A. Chernova, K. Karki, F. Omenya, P. J. Chupas, L. F. J. Piper, M. S. Whittingham, K. W. Chapman, and S. P. Ong, Chem. Mater., 2016, 28, 1794-1805.
- Z. Chen, Q. Chen, L. Chen, R. Zhang, H. Zhou, N. A. Chernova, M. S. Whittingham, J. Electrochem. Soc. 2013, 160, A1777-A1780.
- B. M. Azmi, H. S. Munirah, T. Ishihara, Ionics, 2005, 11.
- M. Bianchini, J. M. Ateba-Mba, P. Dagault, E. Bogdan, D. Carlier, E. Suard, C. Masquelier, L. Croguennec, J. Mater. Chem. A, 2014, 2, 10182-10192.
- K. L. Harrison, C. A. Bridges, C. U. Segre, C. D. Varnado Jr., D. Applestone, C. W. Bielawski, M. P. Paranthaman, A. Manthiram, Chem. Mater., 2014, 26, 3849-3861.

What is claimed is:

1. A cathode composition, comprising ϵ -VOPO₄ nanoparticles in an amount of at least 75% by weight, coated with graphene nanoplatelets in an amount of at least 3% by weight, wherein the graphene nanoplatelets form a conductive network between the ϵ -VOPO₄ nanoparticles.
2. The cathode composition according to claim 1, wherein the ϵ -VOPO₄ nanoparticles are solvothermally or hydrothermally generated.
3. The cathode composition according to claim 1, wherein the graphene nanoplatelets form a layer 10 nm thick on a surface of the ϵ -VOPO₄ nanoparticles.
4. The cathode composition according to claim 1, wherein the ϵ -VOPO₄ nanoparticles are between 100 nm and 200 nm in diameter.
5. The cathode composition according to claim 1, wherein the ϵ -VOPO₄ nanoparticles are intercalated with lithium ions.
6. The cathode composition according to claim 1, wherein the ϵ -VOPO₄ nanoparticles are adapted to intercalate dual lithium ions per vanadium, with an efficiency of at least 80%.
7. The cathode composition according to claim 1, wherein the ϵ -VOPO₄ nanoparticles are adapted to intercalate dual lithium ions per vanadium, with an efficiency of at least 90%.
8. The cathode composition according to claim 1, wherein the graphene nanoplatelets have a surface area of 750 m² per gm.
9. The cathode composition according to claim 1, wherein the graphene nanoplatelets are coated on the ϵ -VOPO₄ nanoparticles by milling.
10. The cathode composition according to claim 1, further comprising a binder in an amount of at least 2% by weight, adapted to bind the graphene nanoplatelet coated ϵ -VOPO₄ nanoparticles together.
11. The cathode composition according to claim 10, wherein the binder comprises polyvinylidene fluoride.
12. A battery comprising:
 - the cathode composition comprising ϵ -VOPO₄ nanoparticles in an amount of at least 75% by weight, coated with graphene nanoplatelets in an amount of at least 3% by weight, wherein the graphene nanoplatelets form a conductive network between the ϵ -VOPO₄ nanoparticles;
 - an electrolyte with a lithium salt;
 - a separator; and
 - an anode.
13. The battery according to claim 12, wherein the anode comprises lithium metal.
14. The battery according to claim 12, having a first voltage plateau region at ~3.8 V and a second voltage plateau region at ~2 V.
15. The battery according to claim 12, having a power density of at least 850 mWh/g.
16. The battery according to claim 12, having an energy density of at least at least 275 mAh/g at a discharge rate of C/20.
17. A cathode for a rechargeable lithium ion battery, comprising:
 - ϵ -VOPO₄ nanoparticles;
 - graphene nanoplatelets coating the ϵ -VOPO₄ nanoparticles to form a conductive network; and
 - a binder adapted to bind the graphene nanoplatelet coated ϵ -VOPO₄ nanoparticles together and maintain the conductive network.

18. The cathode according to claim 17, further comprising a current collector on at least a surface of the cathode, adapted to communicate an electrical current with the conductive network.

19. The cathode according to claim 17, wherein the ϵ -VOPO₄ nanoparticles are solvothermally or hydrothermally generated, the graphene nanoplatelets form a layer 10 nm thick on a surface of the ϵ -VOPO₄ nanoparticles, and the ϵ -VOPO₄ nanoparticles are between 100 nm and 200 nm in diameter.

20. The cathode according to claim 17, wherein at least a portion of the ϵ -VOPO₄ nanoparticles are dual intercalated with lithium ions.

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